THERE IS NO 2-$(15, 5, 2)$ DESIGN: STRUCTURED PROOFS

Abstract. We give the Hall–Connor embedding argument for a hypothetical 2-$(15, 5, 2)$ design, allowing repeated labelled blocks. The resulting integer matrix of order 22 contradicts an elementary determinant obstruction. The proofs follow Lamport's format and record the interfaces and dependencies of the Lean formalization.

Contents

1. The Hall–Connor nonexistence theorem 2
1.1. Definitions and Mathlib inputs 2
1.2. Counts and intersections 4
1.3. Connor's matrix and its rank 7
1.4. Finite certificates and local restrictions 10
1.5. The local six-cliques 13
1.6. The global family 20
1.7. The labelled extension and the matrix it supplies 21
1.8. The determinant obstruction and the conclusion 25
1.9. Interfaces, provenance, and checks 28
References 29

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1. The Hall–Connor nonexistence theorem

We will follow the ordinary proof of `cgss/proofs/hallconnor/hallconnor.tex`. The argument proves the nonexistence assertion used in Remark 5.12 and Example 5.13 of [CGSS], together with its complementary parameters. We use Hall and Connor's argument [HC], including proofs of the instances they take from Connor. We have not consulted Connor's paper directly. We omit the two comparison remarks of the ordinary proof, which the argument does not use.

We follow the blueprint conventions and Lamport [L95]: Assume introduces hypotheses, Prove the conclusion, Let a definition, and Pick a witness. Each numbered step has a proof. A reference $\langle 0\rangle{:}i$ means hypothesis i of the result; $\langle k\rangle$ means the assumption of the enclosing level-k step. Closing Qed steps state why the preceding steps suffice. The tags **[new]** and **[repair]** mark a change of argument and a filled gap, respectively. The no-op commands `\lean` and `\uses` record the Lean declarations and the direct dependencies.

We use the notation of `cgss/notation.md`, Sections 7 and 10: `Design P B v k b r lam`, `D.incMatrix`, and `D.compl`. The Hall–Connor names not listed individually there retain the names of the ordinary proof, following the namespace and naming rules of Section 10. Points and block labels are separate finite types with decidable equality. All sets used in counts are finite sets; their cardinalities are natural numbers. Moment identities and subtractions in weighted sums are evaluated in $\mathbb{Z}$ or $\mathbb{Q}$, never by truncated natural subtraction. Matrix entries are $M(u, w)$. We write I_U for the identity, $J_{U,V}$ for the all-ones matrix, and $\mathbf{j}_U$ for the all-ones vector. Omitted index types are determined by the products. All ranks are over $\mathbb{Q}$.

1.1. Definitions and Mathlib inputs.

Definition (labelled designs).
Assume: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. $v, k, b, r, \lambda \in \mathbb{N}$, with $v > 1$, $k > 1$, and $b > 0$.
A design D with these parameters consists of a function $B \mapsto D(B) \subseteq P$ such that

$$|P| = v, \quad |\mathcal{B}| = b, \quad |D(B)| = k \quad (B \in \mathcal{B}),$$
$$|\{B : p \in D(B)\}| = r \quad (p \in P),$$
$$|\{B : p, q \in D(B)\}| = \lambda \quad (p, q \in P,\ p \neq q).$$

We write $p \in B$ for $p \in D(B)$; the map D need not be injective. The predicate D.IsSymmetric means $v = b$. In Lean the parameter names are `v k b r lam`; any positivity fields of the shared interface are satisfied by the parameters here.

Definition (incidence and block matrices).
Assume: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters (v, k, b, r, λ).
Let: $N \triangleq D$.incMatrix $\in \mathrm{Mat}_{P\times\mathcal{B}}(\mathbb{Z})$, where

$$N(p, B) \triangleq \begin{cases} 1, & p \in B, \\ 0, & p \notin B. \end{cases}$$

Write $N_{\mathbb{Q}}$ for its entrywise rational cast. Put $G \triangleq N_{\mathbb{Q}}^T N_{\mathbb{Q}}$ and $C \triangleq 35I_{\mathcal{B}} - 7G + 10J_{\mathcal{B}}$, both over $\mathbb{Q}$. A product containing G or C uses $N_{\mathbb{Q}}$; we abbreviate it by N in rational calculations. Products asserted integral use the original N.

Definition (the block graph and all six-cliques).
Assume: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters (v, k, b, r, λ).
Let: Z be the simple graph on $\mathcal{B}$ with

$$B \sim B' \iff B \neq B' \text{ and } |B \cap B'| = 1.$$

Here $B \cap B'$ means $D(B) \cap D(B')$. Put $\Gamma(B) \triangleq \{B' : B \sim B'\}$ and

$$\mathcal{S} \triangleq D.\text{sixCliques} \triangleq \{S \subseteq \mathcal{B} : |S| = 6 \text{ and } \forall X, Y \in S,\ X \neq Y \Rightarrow X \sim Y\}.$$

Thus $\mathcal{S}$ is the finite set of *all* six-cliques, not a choice of local cliques. Its members form a finite subtype when used as indices.

Finite counting and order. All declarations below were checked in Mathlib at commit `082e2d37` (4 September 2026), the version pinned in `lake-manifest.json`; paths are relative to `Mathlib/`. Finite double counting is interchange of indicator sums, followed by grouping fibres:

- `Finset.sum_comm`, in `Algebra/BigOperators/Group/Finset/Sigma.lean`;
- `Finset.card_eq_sum_card_fiberwise`, in `Algebra/BigOperators/Group/Finset/Basic.lean`;
- `Finset.card_sdiff_of_subset`, `Finset.card_union_add_card_inter`, `Finset.eq_of_subset_of_card_le`, `Finset.card_pos`, and `Finset.card_le_one`, `Finset.card_eq_one`, `Finset.card_eq_two`, `Finset.card_eq_three`, and `Finset.card_image_of_injective`, in `Data/Finset/Card.lean`;
- `Fintype.bijective_iff_injective_and_card` and `Fintype.equivFin`, in `Data/Fintype/EquivFin.lean`;
- `Fintype.card_sum`, in `Data/Fintype/Sum.lean`, and `Fintype.card_option`, in `Data/Fintype/Option.lean`;
- `Fintype.sum_sum_type` and `Fintype.sum_option`, in `Data/Fintype/BigOperators.lean`;
- `Finset.sum_eq_zero_iff_of_nonneg`, in `Algebra/Order/BigOperators/Group/Finset.lean`.

The last declaration says that a sum of nonnegative terms is zero exactly when every term is zero. The equal-cardinality theorem turns an injection between finite types of equal size into a bijection. An enumeration of a finite subset is an equivalence to its subtype; its inclusion in the ambient type is injective. Integer and rational identities below use ring arithmetic, nonnegative squares, and finite sum distributivity. Small numerical tables are explicit computations.

Matrix operations and reindexing. We use `Matrix.mul_apply`, `Matrix.mul_assoc`, `Matrix.transpose_mul`, `Matrix.dotProduct_mulVec`, `Matrix.mulVec_mulVec`, `Matrix.map_mul`, and `Matrix.submatrix_mul_equiv`, all in `Data/Matrix/Mul.lean`. The last identity says that, for an equivalence e of the summed index type,

$$(A \text{ submatrix } f\, e)(B \text{ submatrix } e\, g) = (AB) \text{ submatrix } f\, g.$$

We also use `Matrix.transpose_map` and `Matrix.transpose_submatrix`, in `LinearAlgebra/Matrix/Defs.lean`, and `Matrix.map_one`, in `Data/Matrix/Diagonal.lean`. Applied to the ring homomorphism $\mathbb{Z} \to \mathbb{Q}$, these transport products, transposes, and I. The cast of J is J entrywise because 1 maps to 1; addition and scalar multiplication commute with the cast entrywise.

Ranks and determinants. The following declarations are in `LinearAlgebra/Matrix/Rank.lean`: `Matrix.rank_self_mul_transpose`, `Matrix.rank_transpose`, `Matrix.rank_of_isUnit`, `Matrix.rank_of_det_ne_zero`, `Matrix.rank_add_rank_le_card_of_mul_eq_zero`, and `Matrix.rank_submatrix_le`. The identity $\operatorname{rk}(NN^T) = \operatorname{rk} N$ applies over the linearly ordered field $\mathbb{Q}$; rank under transpose applies over a field. A nonzero determinant gives full rank. If $AB = 0$, the sum of their ranks is at most the size of the shared index type.

`Matrix.isUnit_iff_isUnit_det`, in `LinearAlgebra/Matrix/NonsingularInverse.lean`, turns a nonzero rational determinant into an invertible matrix.

Determinant arithmetic. We use `Matrix.det_mul`, `Matrix.det_transpose`, `Matrix.det_smul`, `Matrix.det_one`, and `Int.cast_det`, in `LinearAlgebra/Matrix/Determinant/Basic.lean`. We use the matrix determinant lemma `Matrix.det_one_add_replicateCol_mul_replicateRow`, in `LinearAlgebra/Matrix/SchurComplement.lean`, in the form $\det(I + \mathbf{u}\mathbf{v}^T) = 1 + \mathbf{v}^T\mathbf{u}$ (the shared singleton index is `Unit`). These give multiplicativity, scalar homogeneity, and compatibility with the cast $\mathbb{Z} \to \mathbb{Q}$. In particular, the determinant over $\mathbb{Q}$ of an integral matrix is the cast of its integer determinant. We do not use the analytic positive-semidefinite determinant theorem, whose scalar hypotheses do not include $\mathbb{Q}$.
Integer divisibility. We use `Prime.dvd_of_dvd_pow`, in `Algebra/Prime/Defs.lean`, `Int.prime_iff_natAbs_prime`, in `RingTheory/Int/Basic.lean`, and `Nat.prime_five`, in `Data/Nat/Prime/Defs.lean`. The last two prove that 5 is prime in $\mathbb{Z}$; the first gives $5 \mid d^2 \Rightarrow 5 \mid d$ for $d \in \mathbb{Z}$.

1.2. **Counts and intersections.**

Lemma (the forced parameters).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and $|P| = 15$.
2. $F : \mathcal{B} \to \mathrm{Finset}(P)$ has $|F(B)| = 5$ for every B.
3. Each pair of distinct points lies in exactly two labelled blocks of F.
PROVE: Every point is in seven labels, $|\mathcal{B}| = 21$, and

$$\exists D : \texttt{Design P B 15 5 21 7 2},\ \forall B \in \mathcal{B},\ D.\mathrm{block}(B) = F(B).$$

Here the Lean type variable `B` denotes $\mathcal{B}$.

PROOF SKETCH: We will count first through one point, then over all points.
⟨1⟩1. ASSUME: $p \in P$.
PROVE: $r_p \triangleq |\{B : p \in F(B)\}| = 7$.
⟨2⟩1. The set of pairs (q, B) with $q \neq p$ and $p, q \in F(B)$ has size $4r_p$.
PROOF: For each of the r_p labels, deleting p from its five points leaves four choices, by ⟨0⟩:2 and the fibre count in F1. □
⟨2⟩2. The same set has size 28.
PROOF: There are $15 - 1 = 14$ choices of q and two labels for each, by ⟨0⟩:1, ⟨0⟩:3 and F1. □
⟨2⟩3. Q.E.D.
PROOF: The equality $4r_p = 28$ from ⟨2⟩1 and ⟨2⟩2 gives $r_p = 7$. □
⟨1⟩2. $5|\mathcal{B}| = 105$ and $|\mathcal{B}| = 21$.
PROOF: Count (p, B) with $p \in F(B)$ by blocks and by points, using ⟨0⟩:1, ⟨0⟩:2, ⟨1⟩1, and F1: $5|\mathcal{B}| = 15 \cdot 7$. Division by five gives the second assertion. □
⟨1⟩3. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2, with ⟨0⟩:1, ⟨0⟩:2, and ⟨0⟩:3, give every field of Definition (labelled designs). Choose the structure with block function F; its block equality is reflexivity, and $15 > 1$, $5 > 1$, $21 > 0$ are numerical. □

Lemma (incidence identities).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: The integral incidence matrix has

$$NN^T = 5I_P + 2J_P, \quad N\mathbf{j}_{\mathcal{B}} = 7\mathbf{j}_P, \quad N^T\mathbf{j}_P = 5\mathbf{j}_{\mathcal{B}}.$$

Over $\mathbb{Q}$, $G(B, B') = |B \cap B'|$, $G^T = G$, and

$$G^2 = 5G + 50J_{\mathcal{B}}, \quad GJ_{\mathcal{B}} = J_{\mathcal{B}}G = 35J_{\mathcal{B}}, \quad J_{\mathcal{B}}^2 = 21J_{\mathcal{B}}.$$

PROOF SKETCH: The matrix entries count points or labels; associativity then gives the identities for G.

⟨1⟩1. $(NN^T)(p,p) = 7$ for every $p \in P$.
PROOF: By Definitions (incidence and block matrices) and (labelled designs), the entry is the sum of the squares of the seven incidence indicators equal to one; use F2. □

⟨1⟩2. ASSUME: $p, q \in P$ and $p \neq q$.
PROVE: $(NN^T)(p,q) = 2$.
PROOF: The product of the two indicators is one exactly on labels containing p, q, of which there are two by Definition (labelled designs); expand by F2. □

⟨1⟩3. $NN^T = 5I_P + 2J_P$.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2 treat the exhaustive alternatives $p = q$ and $p \neq q$. □

⟨1⟩4. $N\mathbf{j}_{\mathcal{B}} = 7\mathbf{j}_P$.
PROOF: The row sum counts the seven labels on p, by Definition (labelled designs) and F2. □

⟨1⟩5. $N^T\mathbf{j}_P = 5\mathbf{j}_{\mathcal{B}}$.
PROOF: The column sum counts the five points of a block, by Definition (labelled designs) and F2. □

⟨1⟩6. $G(B, B') = |B \cap B'|$ and $G^T = G$.
PROOF: In Definition (incidence and block matrices), the summand $N(p, B)N(p, B')$ is the intersection indicator. Its symmetry in B, B' gives symmetry of G, by F2. □

⟨1⟩7. $G^2 = 5G + 50J_{\mathcal{B}}$.
PROOF: Cast ⟨1⟩3 and ⟨1⟩5 entrywise using the product, transpose, and constant-matrix rules of F2. Then

$$\begin{aligned} G^2 &= N^T(NN^T)N && [\text{F2}] \\ &= 5G + 2N^T J_P N && [\langle 1\rangle 3] \\ &= 5G + 50J_{\mathcal{B}} && [\langle 1\rangle 5,\ 5 \cdot 5 = 25]. \end{aligned}$$

□

⟨1⟩8. $GJ_{\mathcal{B}} = 35J_{\mathcal{B}}$.
PROOF: Steps ⟨1⟩4 and ⟨1⟩5 give $G\mathbf{j}_{\mathcal{B}} = N^T(7\mathbf{j}_P) = 35\mathbf{j}_{\mathcal{B}}$; each column of J is $\mathbf{j}$, by F2. □

⟨1⟩9. $J_{\mathcal{B}}G = 35J_{\mathcal{B}}$.
PROOF: Transpose ⟨1⟩8, using the symmetry in ⟨1⟩6 and F2. □

⟨1⟩10. $J_{\mathcal{B}}^2 = 21J_{\mathcal{B}}$.
PROOF: Each entry sums $|\mathcal{B}| = 21$ ones, by Definition (labelled designs) and F2. □

⟨1⟩11. Q.E.D.
PROOF: Steps ⟨1⟩3–⟨1⟩10 give every asserted identity. □

Lemma (intersection moments).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $B \in \mathcal{B}$ and $n_i \triangleq |\{B' \neq B : |B \cap B'| = i\}|$ for $0 \leq i \leq 5$.
PROVE: Over $\mathbb{Z}$, $\sum_{i=0}^{5} n_i = 20$, $\sum_{i=0}^{5} in_i = 30$, and $\sum_{i=0}^{5} i(i-1)n_i = 20$. In particular $\sum_{B' \neq B} |B \cap B'| = 30$.

PROOF SKETCH: We will count labels, incidences, and ordered pairs of distinct points. The moment statements will be cited outside this proof.

⟨1⟩1. $\sum_{i=0}^{5} n_i = 20$.
PROOF: Intersection size is between zero and five; the fibres partition the $21 - 1$ other labels, by Definition (labelled designs) and F1. □

⟨1⟩2. $\sum_{i=0}^{5} in_i = 30$ and $\sum_{B' \neq B} |B \cap B'| = 30$.

PROOF: Count (p, B') with $p \in B \cap B'$ and $B' \neq B$. For each of five points, six of its seven labels remain. Grouping by B' and then by i gives both sums, using Definition (labelled designs) and F1. □

⟨1⟩3. $\sum_{i=0}^{5} i(i-1)n_i = 20$.
PROOF: Count (p, q, B') with $p, q \in B \cap B'$, $p \neq q$, $B' \neq B$. Each of $5 \cdot 4$ ordered pairs in B has exactly one further label; a label with intersection size i contributes $i(i-1)$, using Definition (labelled designs) and F1. □

⟨1⟩4. Q.E.D.
PROOF: Steps ⟨1⟩1, ⟨1⟩2, and ⟨1⟩3 prove the four assertions. □

Lemma (block intersections). [[HC], Lemma 4.1 at $k = 5$.]
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: Distinct labels meet in one or two points. Every block has exactly ten other labels of each kind; in particular $|\Gamma(B)| = 10$.

PROOF SKETCH: A nonnegative linear combination of the moments excludes intersection sizes other than one and two.

⟨1⟩1. ASSUME: $B \in \mathcal{B}$, with n_i as in Lemma (intersection moments).
PROVE: Every other label meets B in one or two points, and exactly ten labels have each intersection size.

⟨2⟩1. $\sum_{i=0}^{5}(i-1)(i-2)n_i = 0$.
PROOF: The polynomial equals $i(i-1) - 2i + 2$; Lemma (intersection moments) gives $20 - 2 \cdot 30 + 2 \cdot 20 = 0$. □

⟨2⟩2. $n_0 = n_3 = n_4 = n_5 = 0$.
⟨3⟩1. The coefficients $(i-1)(i-2)$, in order $i = 0, 1, 2, 3, 4, 5$, are $2, 0, 0, 2, 6, 12$.
PROOF: Direct integer substitution in the polynomial of ⟨2⟩1. □
⟨3⟩2. Q.E.D.
PROOF: The n_i are nonnegative cardinalities. Apply the zero-sum criterion of F1 to ⟨2⟩1 and ⟨3⟩1; the four indicated coefficients are positive. □

⟨2⟩3. $n_1 = n_2 = 10$.
PROOF: Lemma (intersection moments) and ⟨2⟩2 give $n_1 + n_2 = 20$ and $n_1 + 2n_2 = 30$. Subtract the equations. □

⟨2⟩4. Q.E.D.
PROOF: The zero fibres of ⟨2⟩2 exclude the other intersection sizes; ⟨2⟩3 and Definition (the block graph and all six-cliques) give the multiplicities and degree. □

⟨1⟩2. Q.E.D.
PROOF: Step ⟨1⟩1 applies to every B; Definition (the block graph and all six-cliques) identifies its ten one-point intersections with $\Gamma(B)$. □

Lemma (common-neighbour counts). [[HC], Lemma 4.2 at $k = 5$.]
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $B_1 \neq B_2$ are block labels, $s \triangleq |B_1 \cap B_2|$, and $x_{ij} \triangleq |\{B \notin \{B_1, B_2\} : |B_1 \cap B| = i,\ |B_2 \cap B| = j\}|$.
PROVE: The four counts are

s	x_{11}	x_{12}	x_{21}	x_{22}
1	5	4	4	6
2	4	6	6	3

.

Thus adjacent vertices have five common neighbours and distinct nonadjacent vertices have four.

PROOF SKETCH: We will expose the four filtered-sum equations, including both terms removed from the entry of G^2, before solving them.

⟨1⟩1. The four fibres partition $\mathcal{B} \setminus \{B_1, B_2\}$, and $x_{11} + x_{12} + x_{21} + x_{22} = 19$.
PROOF: Lemma (block intersections) partitions the $21-2$ remaining labels into the four disjoint fibres; use ⟨0⟩:2, F1, and Definition (labelled designs). □

⟨1⟩2. $x_{11} + x_{12} + 2x_{21} + 2x_{22} = 30 - s$.
PROOF: From the last conclusion of Lemma (intersection moments) for B_1, remove the term B_2 of size s. The four fibres have first weights $1, 1, 2, 2$, using ⟨1⟩1, Lemma (block intersections), ⟨0⟩:2, and F1. □

⟨1⟩3. $x_{11} + 2x_{12} + x_{21} + 2x_{22} = 30 - s$.
PROOF: Apply Lemma (intersection moments) to B_2 and remove the term B_1 of size s. The second weights are $1, 2, 1, 2$, using ⟨1⟩1, Lemma (block intersections), ⟨0⟩:2, and F1. □

⟨1⟩4. $x_{11} + 2x_{12} + 2x_{21} + 4x_{22} = 50 - 5s$.
⟨2⟩1. $(G^2)(B_1, B_2) = 5s + 50$.
PROOF: Substitute $G(B_1, B_2) = s$ into the identity $G^2 = 5G + 50J$ of Lemma (incidence identities). □
⟨2⟩2. The terms indexed by B_1 and B_2 in $(G^2)(B_1, B_2)$ are each $5s$.
PROOF: The products are $G(B_1, B_1)G(B_1, B_2)$ and $G(B_1, B_2)G(B_2, B_2)$; both diagonals equal five by Definition (labelled designs) and Lemma (incidence identities). □
⟨2⟩3. Q.E.D.
PROOF: Expand the matrix entry using F2, remove the two distinct terms of ⟨2⟩2 from ⟨2⟩1, then group the remaining terms by their weights ij using ⟨1⟩1, Lemma (block intersections), ⟨0⟩:2, and F1. This gives $(5s + 50) - 10s = 50 - 5s$. □

⟨1⟩5. $x_{22} = 9 - 3s$, $x_{12} = x_{21} = 2 + 2s$, and $x_{11} = 6 - s$.
PROOF: Subtract ⟨1⟩2 and ⟨1⟩3 from ⟨1⟩4, then add ⟨1⟩1, to get $x_{22} = 9 - 3s$. Subtract ⟨1⟩1 and this value from each weighted marginal, then use the total for x_{11}. □

⟨1⟩6. CASE: $s = 1$.
The count vector is $(5, 4, 4, 6)$.
PROOF: Substitute ⟨1⟩ in ⟨1⟩5. □

⟨1⟩7. CASE: $s = 2$.
The count vector is $(4, 6, 6, 3)$.
PROOF: Substitute ⟨1⟩ in ⟨1⟩5. □

⟨1⟩8. $\Gamma(B_1) \cap \Gamma(B_2)$ is exactly the x_{11} fibre; $B_1 \sim B_2$ if and only if $s = 1$.
PROOF: By Definition (the block graph and all six-cliques), membership in both neighbourhoods is intersection size one with each block; irreflexivity excludes B_1, B_2. Conversely the fibre excludes these two labels and gives both adjacencies. For the pair itself, ⟨0⟩:2 and Lemma (block intersections) leave precisely $s = 1$ for adjacency and $s = 2$ for nonadjacency. □

⟨1⟩9. Q.E.D.
PROOF: Lemma (block intersections) makes the cases ⟨1⟩6, ⟨1⟩7 exhaustive. Step ⟨1⟩8 identifies their x_{11} counts with the asserted common-neighbour counts. □

1.3. **Connor's matrix and its rank.**

Lemma (a scalar matrix plus a constant matrix).
ASSUME: 1. U is a finite type with decidable equality, $n = |U| \geq 1$.
2. $\alpha, \beta \in \mathbb{Q}$ and $\alpha \neq 0$.
PROVE: $\det(\alpha I_U + \beta J_U) = \alpha^{n-1}(\alpha + n\beta)$.

PROOF SKETCH: Factor out the nonzero scalar and apply the matrix determinant lemma from F4 to a rank-one product.

⟨1⟩1. $\alpha I + \beta J = \alpha\big(I + (\frac{\beta}{\alpha}\mathbf{j}_U)\mathbf{j}_U^T\big)$.
PROOF: Every entry of $\mathbf{j}_U\mathbf{j}_U^T$ is one. Distribute α and cancel $\alpha \neq 0$ from ⟨0⟩:2. □
⟨1⟩2. **[new]** $\det\big(I + (\frac{\beta}{\alpha}\mathbf{j}_U)\mathbf{j}_U^T\big) = 1 + n\frac{\beta}{\alpha}$.
PROOF: Apply `Matrix.det_one_add_replicateCol_mul_replicateRow` from F4, with singleton index `Unit`, $\mathbf{u} = \frac{\beta}{\alpha}\mathbf{j}_U$, and $\mathbf{v} = \mathbf{j}_U$. The dot product sums n copies of β/α. □
⟨1⟩3. $\det(\alpha I + \beta J) = \alpha^{n-1}(\alpha + n\beta)$.
PROOF: By `Matrix.det_smul` in F4, steps ⟨1⟩1 and ⟨1⟩2 give $\alpha^n(1+n\beta/\alpha)$. Use $n = (n-1)+1$ from ⟨0⟩:1 and cancel $\alpha \neq 0$. □
⟨1⟩4. Q.E.D.
PROOF: Step ⟨1⟩3. □

Lemma (Connor identities and entries).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: Over $\mathbb{Q}$, $C^T = C$, $C^2 = 35C$, $CN^T = 0$, and

$$C(B, B') = \begin{cases} 10, & B = B', \\ 3, & B \sim B', \\ -4, & B \neq B' \text{ and } B \not\sim B'. \end{cases}$$

PROOF SKETCH: We will substitute the identities for G into $C = 35I - 7G + 10J$ and check its three entry types separately.
⟨1⟩1. $C^T = C$.
PROOF: I, J, G are symmetric by Lemma (incidence identities); transpose the defining linear combination in Definition (incidence and block matrices). □
⟨1⟩2. ASSUME: $B = B'$.
PROVE: $C(B, B') = 10$.
PROOF: Lemma (incidence identities) and Definition (labelled designs) give $G(B, B) = 5$, so Definition (incidence and block matrices) gives $35 - 7 \cdot 5 + 10 = 10$. □
⟨1⟩3. ASSUME: $B \sim B'$.
PROVE: $C(B, B') = 3$.
PROOF: Definition (the block graph and all six-cliques) gives distinct labels and intersection size one. Definition (incidence and block matrices) and Lemma (incidence identities) give $0 - 7 \cdot 1 + 10 = 3$. □
⟨1⟩4. ASSUME: $B \neq B'$ and $B \not\sim B'$.
PROVE: $C(B, B') = -4$.
PROOF: Lemma (block intersections) and Definition (the block graph and all six-cliques) force intersection size two. Definition (incidence and block matrices) gives $0 - 7 \cdot 2 + 10 = -4$. □
⟨1⟩5. $C^2 = 35C$.
PROOF: In the following calculation I, G, J are indexed by $\mathcal{B}$:

$$\begin{aligned} C^2 &= 1225I + 49G^2 + 100J^2 - 490G + 700J - 140GJ && \text{[(incidence identities), } GJ = JG\text{]} \\ &= 1225I + 49(5G + 50J) - 490G - 2100J && \text{[(incidence identities), arithmetic]} \\ &= 1225I - 245G + 350J && \text{[ring arithmetic]} \\ &= 35C && \text{[(incidence and block matrices)].} \end{aligned}$$

□
⟨1⟩6. $CN^T = 0$.
⟨2⟩1. $N^T J_P = 5J_{\mathcal{B},P}$ and $J_{\mathcal{B}}N^T = 7J_{\mathcal{B},P}$.
PROOF: Each entry in the first product is a column sum of N, and each in the second a row sum; apply Lemma (incidence identities) and F2. □

⟨2⟩2. Q.E.D.
PROOF: By associativity from F2,

$$\begin{aligned} CN^T &= 35N^T - 7N^T(NN^T) + 10J_{\mathcal{B}}N^T && \text{[(incidence and block matrices)]} \\ &= 35N^T - 7N^T(5I_P + 2J_P) + 70J_{\mathcal{B},P} && \text{[(incidence identities), ⟨2⟩1]} \\ &= 0 && \text{[⟨2⟩1, ring arithmetic].} \end{aligned}$$

□

⟨1⟩7. Q.E.D.
PROOF: Steps ⟨1⟩2, ⟨1⟩3, ⟨1⟩4 cover every pair of indices. Steps ⟨1⟩1, ⟨1⟩5, ⟨1⟩6 give the other conclusions. □

Lemma (quadratic inequality and rank bound).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: For every $\mathbf{x} \in \mathbb{Q}^{\mathcal{B}}$, $35\mathbf{x}^T C\mathbf{x} = (C\mathbf{x})^T(C\mathbf{x}) \geq 0$, so $\mathbf{x}^T C\mathbf{x} \geq 0$; also $\operatorname{rk} C \leq 6$.

PROOF SKETCH: The equation $C^2 = 35C$ gives a sum of squares. The equation $CN^T = 0$ gives the rank bound after proving N has rank fifteen.
⟨1⟩1. For every $\mathbf{x}$, $35\mathbf{x}^T C\mathbf{x} = (C\mathbf{x})^T(C\mathbf{x}) \geq 0$.
PROOF: Lemma (Connor identities and entries) gives $35C = C^T C$. Multiply by $\mathbf{x}^T, \mathbf{x}$, using F2; the resulting value is a sum of rational squares. □
⟨1⟩2. For every $\mathbf{x}$, $\mathbf{x}^T C\mathbf{x} \geq 0$.
PROOF: Divide ⟨1⟩1 by $35 > 0$. □
⟨1⟩3. $\det(NN^T) = 5^{14} \cdot 35 \neq 0$.
PROOF: First cast Lemma (incidence identities) to $\mathbb{Q}$ by F2. Apply Lemma (a scalar matrix plus a constant matrix). Definition (labelled designs) gives $|P| = 15$; take $\alpha = 5 \neq 0$ and $\beta = 2$. □
⟨1⟩4. $\operatorname{rk} N = \operatorname{rk} N^T = 15$.
⟨2⟩1. $\operatorname{rk}(NN^T) = 15$.
PROOF: The determinant in ⟨1⟩3 is a unit in $\mathbb{Q}$. The unit determinant and full-rank criteria of F3 give rank $|P| = 15$. □
⟨2⟩2. Q.E.D.
PROOF: Apply the two transpose-rank identities of F3 over $\mathbb{Q}$ to ⟨2⟩1. □
⟨1⟩5. $\operatorname{rk} C \leq 6$.
PROOF: By $CN^T = 0$ in Lemma (Connor identities and entries) and the zero-product inequality of F3, $\operatorname{rk} C + \operatorname{rk} N^T \leq |\mathcal{B}| = 21$. Substitute ⟨1⟩4 and subtract fifteen. □
⟨1⟩6. Q.E.D.
PROOF: Steps ⟨1⟩1, ⟨1⟩2, and ⟨1⟩5. □

Lemma (principal matrices on labelled subsets).
ASSUME: 1. $P, \mathcal{B}, U$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $e : U \to \mathcal{B}$ is injective, and $C_e(u, w) \triangleq C(e(u), e(w))$.
PROVE: For all $\mathbf{x} \in \mathbb{Q}^U$, $\mathbf{x}^T C_e\mathbf{x} \geq 0$; $\operatorname{rk} C_e \leq 6$; if $|U| \geq 7$, then $\det C_e = 0$.

PROOF SKETCH: We will extend vectors by zero along the injection. Rank monotonicity also applies to the specified submatrix, so later enumerations need no unstated transport of singularity.
⟨1⟩1. For $\mathbf{x} \in \mathbb{Q}^U$, define $\widehat{\mathbf{x}}(e(u)) = \mathbf{x}(u)$ and $\widehat{\mathbf{x}}(B) = 0$ off the image. This defines $\widehat{\mathbf{x}} \in \mathbb{Q}^{\mathcal{B}}$.
PROOF: Injectivity in ⟨0⟩:2 gives at most one preimage; the two domains partition $\mathcal{B}$. □
⟨1⟩2. $\widehat{\mathbf{x}}^T C\widehat{\mathbf{x}} = \mathbf{x}^T C_e\mathbf{x} \geq 0$.

PROOF: Expand the quadratic sum using F2. Terms off the image vanish by ⟨1⟩1; change both finite indices along e using F1. Apply Lemma (quadratic inequality and rank bound) to $\widehat{\mathbf{x}}$. □

⟨1⟩3. $\operatorname{rk} C_e \le 6$.

PROOF: Rank monotonicity for submatrices in F3 bounds this rank by $\operatorname{rk} C$; use Lemma (quadratic inequality and rank bound). □

⟨1⟩4. ASSUME: $|U| \ge 7$.
PROVE: $\det C_e = 0$.

PROOF: If the determinant were nonzero, the full-rank criterion in F3 would give $\operatorname{rk} C_e = |U| \ge 7$, contradicting ⟨1⟩3. □

⟨1⟩5. Q.E.D.

PROOF: Steps ⟨1⟩2, ⟨1⟩3, and ⟨1⟩4. □

1.4. **Finite certificates and local restrictions.**

Definition (the C19 matrices). The following matrices have integer entries; their rational casts are denoted by the same letters. Indices in the displays run from 1 to the indicated order, corresponding to Lean's $\mathrm{Fin}\, n$ in increasing order.

$$Q_{\mathrm{claw}} = \begin{pmatrix} 10 & 3 & 3 & 3 \\ 3 & 10 & -4 & -4 \\ 3 & -4 & 10 & -4 \\ 3 & -4 & -4 & 10 \end{pmatrix}$$

$$Q_{\mathrm{five}} = \begin{pmatrix} 10 & 3 & 3 & 3 & -4 \\ 3 & 10 & 3 & -4 & 3 \\ 3 & 3 & 10 & -4 & 3 \\ 3 & -4 & -4 & 10 & 3 \\ -4 & 3 & 3 & 3 & 10 \end{pmatrix}$$

$$Q_1 = \begin{pmatrix} 10 & 3 & 3 & 3 & -4 & -4 & -4 \\ 3 & 10 & 3 & -4 & 3 & 3 & 3 \\ 3 & 3 & 10 & -4 & 3 & 3 & 3 \\ 3 & -4 & -4 & 10 & -4 & -4 & -4 \\ -4 & 3 & 3 & -4 & 10 & 3 & 3 \\ -4 & 3 & 3 & -4 & 3 & 10 & 3 \\ -4 & 3 & 3 & -4 & 3 & 3 & 10 \end{pmatrix}$$

$$Q_2 = \begin{pmatrix} 10 & 3 & 3 & 3 & 3 & -4 & -4 \\ 3 & 10 & 3 & -4 & -4 & 3 & 3 \\ 3 & 3 & 10 & -4 & -4 & 3 & 3 \\ 3 & -4 & -4 & 10 & 3 & -4 & -4 \\ 3 & -4 & -4 & 3 & 10 & -4 & -4 \\ -4 & 3 & 3 & -4 & -4 & 10 & 3 \\ -4 & 3 & 3 & -4 & -4 & 3 & 10 \end{pmatrix}$$

$$K_1 = \begin{pmatrix} 20 & -12 & -12 & -6 & 8 & 8 & 8 \\ -12 & 9 & 7 & 4 & -5 & -5 & -5 \\ -12 & 7 & 9 & 4 & -5 & -5 & -5 \\ -6 & 4 & 4 & 4 & -2 & -2 & -2 \\ 8 & -5 & -5 & -2 & 5 & 3 & 3 \\ 8 & -5 & -5 & -2 & 3 & 5 & 3 \\ 8 & -5 & -5 & -2 & 3 & 3 & 5 \end{pmatrix}$$

$$K_2 = \begin{pmatrix} 17 & -11 & -11 & -7 & -7 & 6 & 6 \\ -11 & 9 & 7 & 5 & 5 & -4 & -4 \\ -11 & 7 & 9 & 5 & 5 & -4 & -4 \\ -7 & 5 & 5 & 5 & 3 & -2 & -2 \\ -7 & 5 & 5 & 3 & 5 & -2 & -2 \\ 6 & -4 & -4 & -2 & -2 & 4 & 2 \\ 6 & -4 & -4 & -2 & -2 & 2 & 4 \end{pmatrix}$$

Computation (C19: quadratic values and inverse certificates).
LET: $\mathbf{x} \triangleq (-1,1,1,1)^T$ and $\mathbf{y} \triangleq (-1,1,1,1,-1)^T$.
PROVE: $\mathbf{x}^T Q_{\text{claw}} \mathbf{x} = -2$, $\mathbf{y}^T Q_{\text{five}} \mathbf{y} = -4$, and $K_j Q_j = 14 I_7$ for each $j \in \{1,2\}$.

PROOF SKETCH: These are finite integer certificates, suitable for kernel-checked arithmetic. The script `cgss/proofs/hallconnor/checks/check_proof.py` reads every matrix of Definition (the C19 matrices) from the ordinary proof, where the matrices are the same, and recomputes every entry.

⟨1⟩1. $\mathbf{x}^T Q_{\text{claw}} \mathbf{x} = -2$.
PROOF: From Definition (the C19 matrices), the diagonal sum is 40 and the upper-triangle sum is $-9-12=-21$. The quadratic sum is $40+2(-21)=-2$, by F2. □

⟨1⟩2. $\mathbf{y}^T Q_{\text{five}} \mathbf{y} = -4$.
PROOF: From Definition (the C19 matrices), the diagonal sum is 50. The upper-triangle row contributions are $-13,-4,-7,-3$, totalling -27. Thus the value is $50+2(-27)=-4$, by F2. □

⟨1⟩3. $K_1 Q_1 = 14 I_7$.
PROOF: For all $u, w \in \{1,\dots,7\}$, substitute the integer entries in Definition (the C19 matrices) into $\sum_{i=1}^{7} K_1(u,i) Q_1(i,w)$, using F2. The seven diagonal sums are 14 and all 42 other sums are zero. For example the first row starts with $200-36-36-18-32-32-32=14$ and $60-120-36+24+24+24+24=0$. □

⟨1⟩4. $K_2 Q_2 = 14 I_7$.
PROOF: For all $u, w \in \{1,\dots,7\}$, the sums $\sum_{i=1}^{7} K_2(u,i) Q_2(i,w)$ from Definition (the C19 matrices) are 14 on the diagonal and zero off it, by integer multiplication and F2. The first two sums are $170-33-33-21-21-24-24=14$ and $51-110-33+28+28+18+18=0$. □

⟨1⟩5. Q.E.D.
PROOF: Steps ⟨1⟩1, ⟨1⟩2, ⟨1⟩3, and ⟨1⟩4. □

Lemma (the two nonsingular matrices).
ASSUME: $j \in \{1,2\}$, with Q_j, K_j as in Definition (the C19 matrices).
PROVE: $\det Q_j \neq 0$ over $\mathbb{Q}$.

PROOF SKETCH: The certified product has nonzero determinant.

⟨1⟩1. **[new]** $(\det K_j)(\det Q_j) = 14^7$.
PROOF: Cast the integral identity of Computation (C19: quadratic values and inverse certificates) to $\mathbb{Q}$ by F2, then take determinants; multiplicativity and scalar multiplication in F4 give $\det(14 I_7) = 14^7$. This replaces the ordinary proof's zero-kernel argument with the determinant of the same certificate. □

⟨1⟩2. Q.E.D.
PROOF: If $\det Q_j = 0$, the left side of ⟨1⟩1 vanishes, whereas $14^7 \neq 0$ in $\mathbb{Q}$. □

Lemma (no induced claw).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15,5,21,7,2)$.

PROVE: There are no four distinct blocks B, X, Y, W for which $B \sim X$, $B \sim Y$, $B \sim W$, and X, Y, W are pairwise nonadjacent.

PROOF SKETCH: The four specified labels would give a negative quadratic value of a principal matrix of C.

⟨1⟩1. ASSUME: Four blocks B, X, Y, W satisfy the forbidden pattern.
PROVE: FALSE.

⟨2⟩1. The ordered list (B, X, Y, W) defines an injection $e : \mathrm{Fin}\, 4 \to \mathcal{B}$.
PROOF: The blocks are distinct by ⟨1⟩. □

⟨2⟩2. $C_e(i, i) = 10$ for each i.
PROOF: The diagonal clause of Lemma (Connor identities and entries). □

⟨2⟩3. All entries from the centre to a leaf are 3.
PROOF: The three adjacencies in ⟨1⟩ and the adjacency clause of Lemma (Connor identities and entries). □

⟨2⟩4. All entries between distinct leaves are -4.
PROOF: Their distinctness and nonadjacency in ⟨1⟩ give the last clause of Lemma (Connor identities and entries). □

⟨2⟩5. $C_e = Q_{\mathrm{claw}}$.
PROOF: The diagonal, centre-leaf, and leaf-leaf entries are exactly those of Definition (the C19 matrices), by ⟨2⟩2, ⟨2⟩3, ⟨2⟩4. Symmetry in Lemma (Connor identities and entries) supplies the reversed entries. □

⟨2⟩6. Q.E.D.
PROOF: Apply Lemma (principal matrices on labelled subsets) with ⟨2⟩1 to the vector of Computation (C19: quadratic values and inverse certificates). Step ⟨2⟩5 makes its value $-2 \geq 0$, a contradiction. □

⟨1⟩2. Q.E.D.
PROOF: Step ⟨1⟩1 negates the forbidden existence. □

Lemma (the five-block restriction). [[HC], Lemma 4.5 at $k = 5$.]
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. B_1, B_2, B_3, Y, W are five distinct blocks; B_1, B_2, B_3 form a triangle.
3. $Y \sim B_1$, $Y \not\sim B_2, B_3$, and $W \sim B_2, B_3$, $W \not\sim B_1$.
PROVE: $Y \not\sim W$.

PROOF SKETCH: An edge YW gives the second negative quadratic certificate.

⟨1⟩1. ASSUME: $Y \sim W$.
PROVE: FALSE.

⟨2⟩1. The ordered list (B_1, B_2, B_3, Y, W) defines an injection $e : \mathrm{Fin}\, 5 \to \mathcal{B}$.
PROOF: Distinctness is ⟨0⟩:2. □

⟨2⟩2. Every diagonal entry of C_e is 10.
PROOF: Lemma (Connor identities and entries). □

⟨2⟩3. The three entries between B_1, B_2, B_3 are 3.
PROOF: The triangle in ⟨0⟩:2 and Lemma (Connor identities and entries). □

⟨2⟩4. The entries from Y to (B_1, B_2, B_3) are $(3, -4, -4)$.
PROOF: Apply the adjacency and nonadjacency clauses of Lemma (Connor identities and entries) to the three stipulated relations in ⟨0⟩:3; their labels are distinct by ⟨0⟩:2. □

⟨2⟩5. The entries from W to (B_1, B_2, B_3) are $(-4, 3, 3)$.
PROOF: Apply Lemma (Connor identities and entries) to the three stipulated relations of W in ⟨0⟩:3, using ⟨0⟩:2. □

⟨2⟩6. $C(Y, W) = 3$.
PROOF: By ⟨1⟩ and Lemma (Connor identities and entries). □
⟨2⟩7. $C_e = Q_{\text{five}}$.
PROOF: Steps ⟨2⟩2–⟨2⟩6 give every diagonal and upper-triangle entry in Definition (the C19 matrices); symmetry from Lemma (Connor identities and entries) supplies the lower triangle. □
⟨2⟩8. Q.E.D.
PROOF: Lemma (principal matrices on labelled subsets), ⟨2⟩1, ⟨2⟩7, and Computation (C19: quadratic values and inverse certificates) give $-4 \geq 0$, a contradiction. □
⟨1⟩2. Q.E.D.
PROOF: Step ⟨1⟩1 proves the negation. □

1.5. **The local six-cliques.**

Lemma (splitting a neighbourhood).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $B_1, B_2 \in \mathcal{B}$ and $B_1 \sim B_2$.
PROVE: There are $S_1, S_2 \in \mathcal{S}$ such that

$$B_1, B_2 \in S_1, \quad S_1 \cap S_2 = \{B_1\}, \quad S_1 \cup S_2 = \{B_1\} \cup \Gamma(B_1).$$

Moreover every $K \in \mathcal{S}$ containing B_1, B_2 equals S_1.

PROOF SKETCH: We will partition the other blocks by adjacency to B_1, B_2. The claw restriction gives two cliques and an exact edge count. A suitable third vertex then has a defect $j \leq 2$. The two positive defects give the certified nonsingular seven-block matrices, in conflict with the rank bound. Defect zero gives the split and uniqueness.
LET: $\mathcal{M} \triangleq \Gamma(B_1) \cap \Gamma(B_2)$, $\mathcal{U} \triangleq \Gamma(B_1) \setminus (\Gamma(B_2) \cup \{B_2\})$, $\mathcal{V} \triangleq \Gamma(B_2) \setminus (\Gamma(B_1) \cup \{B_1\})$, and $\mathcal{R} \triangleq \mathcal{B} \setminus (\{B_1, B_2\} \cup \mathcal{M} \cup \mathcal{U} \cup \mathcal{V})$.
Write $\mathcal{F}_{ij} \triangleq \{B \notin \{B_1, B_2\} : |B_1 \cap B| = i,\ |B_2 \cap B| = j\}$ for the fibres whose cardinalities are x_{ij} in Lemma (common-neighbour counts).
⟨1⟩1. $\mathcal{M} = \mathcal{F}_{11}$.
PROOF: Definition (the block graph and all six-cliques) identifies each adjacency with a one-point intersection and excludes the two endpoints by irreflexivity. Conversely the fibre gives both adjacencies. Lemma (block intersections) places these intersections among the two possible sizes for distinct labels. □
⟨1⟩2. $\mathcal{U} = \mathcal{F}_{12}$.
PROOF: Membership in $\mathcal{U}$ gives adjacency to B_1 and excludes B_2 and adjacency to B_2. Definition (the block graph and all six-cliques) gives the first intersection size one; Lemma (block intersections) forces the second to be two. Conversely these two sizes and the exclusions in $\mathcal{F}_{12}$ give precisely the defining membership conditions of $\mathcal{U}$. □
⟨1⟩3. $\mathcal{V} = \mathcal{F}_{21}$.
PROOF: Membership in $\mathcal{V}$ gives adjacency to B_2 and excludes B_1 and adjacency to B_1. Definition (the block graph and all six-cliques) gives the second size one; Lemma (block intersections) forces the first to be two. Conversely the two sizes and endpoint exclusions in $\mathcal{F}_{21}$ give precisely the defining conditions of $\mathcal{V}$. □
⟨1⟩4. $\mathcal{R} = \mathcal{F}_{22}$.
PROOF: Outside $\{B_1, B_2\}$, Lemma (block intersections) leaves four possible pairs of sizes. Steps ⟨1⟩1–⟨1⟩3 remove the first three, leaving $(2, 2)$. Conversely this pair gives no adjacency to either endpoint by Definition (the block graph and all six-cliques), so its label lies in none of $\mathcal{M}, \mathcal{U}, \mathcal{V}$ and belongs to $\mathcal{R}$. □

⟨1⟩5. $|\mathcal{M}| = 5$, $|\mathcal{U}| = 4$, $|\mathcal{V}| = 4$, and $|\mathcal{R}| = 6$.
PROOF: Hypothesis ⟨0⟩:2 and Definition (the block graph and all six-cliques) give $s = 1$. Apply the $s = 1$ row of Lemma (common-neighbour counts) to steps ⟨1⟩1–⟨1⟩4. □
⟨1⟩6. $\mathcal{M}, \mathcal{U}, \mathcal{V}, \mathcal{R}$ are pairwise disjoint. Each is disjoint from $\{B_1, B_2\}$.
PROOF: Their four different pairs of intersection sizes in steps ⟨1⟩1–⟨1⟩4 are mutually exclusive, and each fibre excludes the endpoints. □
⟨1⟩7. $\mathcal{M} \cup \mathcal{U} \cup \mathcal{V} \cup \mathcal{R} = \mathcal{B} \setminus \{B_1, B_2\}$.
PROOF: By Lemma (block intersections), every label outside the endpoints has one of the four pairs of sizes. Use steps ⟨1⟩1–⟨1⟩4. □
⟨1⟩8. $\Gamma(B_1) = \{B_2\} \cup \mathcal{M} \cup \mathcal{U}$, with pairwise disjoint parts.
PROOF: The endpoint B_2 is a neighbour by ⟨0⟩:2. Every other neighbour has first size one, so lies in $\mathcal{F}_{11}$ or $\mathcal{F}_{12}$ by Lemma (block intersections) and Definition (the block graph and all six-cliques); both fibres consist of neighbours. Apply ⟨1⟩1, ⟨1⟩2, and ⟨1⟩6. □
⟨1⟩9. $\Gamma(B_2) = \{B_1\} \cup \mathcal{M} \cup \mathcal{V}$, with pairwise disjoint parts.
PROOF: The endpoint B_1 is a neighbour by ⟨0⟩:2. Every other neighbour has second size one, so lies in $\mathcal{F}_{11}$ or $\mathcal{F}_{21}$ by Lemma (block intersections) and Definition (the block graph and all six-cliques); both fibres consist of neighbours. Apply ⟨1⟩1, ⟨1⟩3, and ⟨1⟩6. □
⟨1⟩10. $\mathcal{U}$ is a clique.
⟨2⟩1. ASSUME: $Y, Y' \in \mathcal{U}$, $Y \neq Y'$, and $Y \not\sim Y'$.
PROVE: FALSE.
PROOF: In ⟨1⟩2, B_2, Y, Y' are distinct neighbours of B_1. Neither Y nor Y' is adjacent to B_2; ⟨2⟩ excludes their mutual edge. This is the claw forbidden by Lemma (no induced claw). □
⟨2⟩2. Q.E.D.
PROOF: Step ⟨2⟩1 excludes every missing edge between distinct members, as required by Definition (the block graph and all six-cliques). □
⟨1⟩11. $\mathcal{V}$ is a clique.
⟨2⟩1. ASSUME: $W, W' \in \mathcal{V}$, $W \neq W'$, and $W \not\sim W'$.
PROVE: FALSE.
PROOF: Step ⟨1⟩3 makes B_1, W, W' distinct neighbours of B_2; B_1 is adjacent to neither W nor W'. With ⟨2⟩ these are the three leaves of a claw, contradicting Lemma (no induced claw). □
⟨2⟩2. Q.E.D.
PROOF: Step ⟨2⟩1 proves the clique predicate of Definition (the block graph and all six-cliques). □
⟨1⟩12. For every $Y \in \mathcal{U}$, $|\Gamma(Y) \cap \mathcal{M}| = 2$.
⟨2⟩1. For each $Y \in \mathcal{U}$, the common neighbours of B_1, Y are the disjoint union of $\mathcal{U} \setminus \{Y\}$ and $\Gamma(Y) \cap \mathcal{M}$.
PROOF: In the first neighbourhood decomposition of ⟨1⟩8, $B_2 \not\sim Y$ by membership in $\mathcal{U}$; step ⟨1⟩10 gives every member of $\mathcal{U} \setminus \{Y\}$. These leave precisely the stated part of $\mathcal{M}$. □
⟨2⟩2. Q.E.D.
PROOF: Lemma (common-neighbour counts) gives five common neighbours of the adjacent pair B_1, Y. The first part in ⟨2⟩1 has size $4 - 1 = 3$ by ⟨1⟩5 and F1, leaving two. □
⟨1⟩13. $\sum_{X \in \mathcal{M}} |\Gamma(X) \cap \mathcal{U}| = \sum_{Y \in \mathcal{U}} |\Gamma(Y) \cap \mathcal{M}|$.
PROOF: Write each cardinality as a sum of adjacency indicators. Interchange the finite sums by `Finset.sum_comm` in F1; symmetry of adjacency from Definition (the block graph and all six-cliques) gives the other sum. □
⟨1⟩14. $\sum_{X \in \mathcal{M}} |\Gamma(X) \cap \mathcal{U}| = 8$.

PROOF: By ⟨1⟩13 the sum equals $\sum_{Y\in\mathcal{U}} 2$, using ⟨1⟩12. Step ⟨1⟩5 gives $4 \cdot 2 = 8$. □

⟨1⟩15. PICK $B_3 \in \mathcal{M}$ such that $|\Gamma(B_3) \cap \mathcal{U}| \geq 2$.

PROOF: Otherwise each summand of ⟨1⟩14 is at most one. Termwise summation gives $8 \leq \sum_{X\in\mathcal{M}} 1 = |\mathcal{M}| = 5$, using ⟨1⟩5 and F1, a contradiction. □

⟨1⟩16. LET: $j \triangleq |\mathcal{U} \setminus \Gamma(B_3)|$, and

$$h \triangleq |\Gamma(B_3) \cap (\mathcal{M} \setminus \{B_3\})|, \quad e \triangleq |\Gamma(B_3) \cap \mathcal{V}|, \quad \ell \triangleq |\Gamma(B_3) \cap \mathcal{R}|.$$

Then $0 \leq j \leq 2$ and $|\Gamma(B_3) \cap \mathcal{U}| = 4 - j$.

PROOF: The neighbours and nonneighbours of B_3 partition the four-element set $\mathcal{U}$, by ⟨1⟩5 and F1; ⟨1⟩15 supplies at least two neighbours. □

⟨1⟩17. $h = j$, $e = 4 - j$, and $\ell = j$.

⟨2⟩1. $1 + h + (4 - j) = 5$.

PROOF: Count common neighbours of B_1, B_3 using Lemma (common-neighbour counts) and ⟨1⟩8: B_2 contributes one, $\mathcal{M} \setminus \{B_3\}$ contributes h, and $\mathcal{U}$ contributes $4 - j$ by ⟨1⟩16. □

⟨2⟩2. $1 + h + e = 5$.

PROOF: Count common neighbours of B_2, B_3 using Lemma (common-neighbour counts): B_1 contributes one, the part $\mathcal{M} \setminus \{B_3\}$ contributes h, and $\mathcal{V}$ contributes e, by ⟨1⟩9 and ⟨1⟩16. □

⟨2⟩3. $2 + h + (4 - j) + e + \ell = 10$.

PROOF: Count all neighbours of B_3 in the partition ⟨1⟩7, whose parts are disjoint by ⟨1⟩6. Both B_1, B_2 are neighbours by ⟨1⟩15; the other contributions are the four counts in ⟨1⟩16. The degree is ten by Lemma (block intersections). □

⟨2⟩4. Q.E.D.

PROOF: Solve ⟨2⟩1 for h, then ⟨2⟩2 for e, then ⟨2⟩3 for ℓ, over $\mathbb{Z}$. □

⟨1⟩18. LET: $\mathcal{U}_0 \triangleq \mathcal{U} \setminus \Gamma(B_3)$ and $\mathcal{V}_1 \triangleq \mathcal{V} \cap \Gamma(B_3)$.

They are disjoint cliques of sizes $j, 4 - j$, disjoint also from $\{B_1, B_2, B_3\}$.

PROOF: Disjointness is ⟨1⟩6 and ⟨1⟩15; the sizes are ⟨1⟩16 and ⟨1⟩17. They are subsets of the cliques in ⟨1⟩10 and ⟨1⟩11. □

⟨1⟩19. No member of $\mathcal{U}_0$ is adjacent to a member of $\mathcal{V}_1$.

⟨2⟩1. ASSUME: $Y \in \mathcal{U}_0$ and $W \in \mathcal{V}_1$.

PROVE: $Y \not\sim W$.

⟨3⟩1. B_1, B_2, B_3, Y, W are distinct.

PROOF: Step ⟨1⟩18 gives disjointness, and ⟨1⟩6, ⟨1⟩15 separate B_3 from the adjacent pair B_1, B_2. □

⟨3⟩2. B_1, B_2, B_3 form a triangle.

PROOF: Use ⟨0⟩:2 and $B_3 \in \mathcal{M}$ in ⟨1⟩15. □

⟨3⟩3. $Y \sim B_1$, $Y \not\sim B_2$, and $Y \not\sim B_3$.

PROOF: Membership in $\mathcal{U}_0$ in ⟨2⟩, and the definition in ⟨1⟩18 and the fibre description in ⟨1⟩2. □

⟨3⟩4. $W \sim B_2$, $W \sim B_3$, and $W \not\sim B_1$.

PROOF: Membership in $\mathcal{V}_1$ in ⟨2⟩, and the definition in ⟨1⟩18 and the fibre description in ⟨1⟩3. □

⟨3⟩5. Q.E.D.

PROOF: Apply Lemma (the five-block restriction) to ⟨3⟩1–⟨3⟩4. □

⟨2⟩2. Q.E.D.

PROOF: Step ⟨2⟩1 is quantified over every pair of the indicated parts. □

⟨1⟩20. On the named blocks B_1, B_2, B_3 and the members of $\mathcal{U}_0, \mathcal{V}_1$, C is symmetric with diagonal 10. Its off-diagonal entries are

	B_1	B_2	B_3	$\mathcal{U}_0$	$\mathcal{V}_1$
B_1	$*$	3	3	3	-4
B_2	3	$*$	3	-4	3
B_3	3	3	$*$	-4	3
$\mathcal{U}_0$	3	-4	-4	3	-4
$\mathcal{V}_1$	-4	3	3	-4	3

A set in this table denotes any of its members, with distinct blocks required for within-set entries.

⟨2⟩1. For every block X, $C(X, X) = 10$.
PROOF: Lemma (Connor identities and entries). □

⟨2⟩2. The three entries among B_1, B_2, B_3 are 3.
PROOF: The triangle follows from ⟨0⟩:2, ⟨1⟩1, and ⟨1⟩15; use Lemma (Connor identities and entries). □

⟨2⟩3. For $Y \in \mathcal{U}_0$, $C(B_1, Y) = 3$.
PROOF: Membership in $\mathcal{U}$ gives adjacency to B_1 by ⟨1⟩2 and ⟨1⟩18; apply Lemma (Connor identities and entries). □

⟨2⟩4. For $Y \in \mathcal{U}_0$, $C(B_2, Y) = -4$.
PROOF: Step ⟨1⟩2 excludes adjacency to B_2; distinctness in ⟨1⟩18 permits Lemma (Connor identities and entries). □

⟨2⟩5. For $Y \in \mathcal{U}_0$, $C(B_3, Y) = -4$.
PROOF: The definition and disjointness in ⟨1⟩18 give nonadjacency and distinctness; apply Lemma (Connor identities and entries). □

⟨2⟩6. For $W \in \mathcal{V}_1$, $C(B_1, W) = -4$.
PROOF: Step ⟨1⟩3 excludes adjacency to B_1; use the distinctness in ⟨1⟩18 and Lemma (Connor identities and entries). □

⟨2⟩7. For $W \in \mathcal{V}_1$, $C(B_2, W) = 3$.
PROOF: Step ⟨1⟩3 gives adjacency to B_2 for every member of the subset in ⟨1⟩18; use Lemma (Connor identities and entries). □

⟨2⟩8. For $W \in \mathcal{V}_1$, $C(B_3, W) = 3$.
PROOF: The definition of $\mathcal{V}_1$ in ⟨1⟩18 gives adjacency; use Lemma (Connor identities and entries). □

⟨2⟩9. For distinct $Y, Y' \in \mathcal{U}_0$, $C(Y, Y') = 3$.
PROOF: The clique in ⟨1⟩18 gives adjacency, so Lemma (Connor identities and entries) applies. □

⟨2⟩10. For distinct $W, W' \in \mathcal{V}_1$, $C(W, W') = 3$.
PROOF: The clique in ⟨1⟩18 gives adjacency; use Lemma (Connor identities and entries). □

⟨2⟩11. For $Y \in \mathcal{U}_0$ and $W \in \mathcal{V}_1$, $C(Y, W) = -4$.
PROOF: Step ⟨1⟩19 excludes the edge; ⟨1⟩18 gives distinctness, so Lemma (Connor identities and entries) applies. □

⟨2⟩12. Q.E.D.
PROOF: Steps ⟨2⟩1–⟨2⟩11 cover the diagonal and all ten unordered pair types. Symmetry from Lemma (Connor identities and entries) gives the reversed entries. □

⟨1⟩21. ASSUME: $j = 1$.
PROVE: FALSE.

⟨2⟩1. PICK Y, W_1, W_2, W_3 with $\mathcal{U}_0 = \{Y\}$, $\mathcal{V}_1 = \{W_1, W_2, W_3\}$, and the W_i pairwise distinct.
PROOF: By ⟨1⟩ and ⟨1⟩18, the sizes are one and three. Apply `Finset.card_eq_one` and `Finset.card_eq_three` from F1. □

⟨2⟩2. Put $a \triangleq (B_1, B_2, B_3, Y, W_1, W_2, W_3) : \mathrm{Fin}\, 7 \to \mathcal{B}$. Then a is injective.
PROOF: The first three blocks are distinct by ⟨0⟩:2, ⟨1⟩6, and ⟨1⟩15. Step ⟨1⟩18 separates them from the two disjoint parts, and ⟨2⟩1 separates the W_i. Thus the literal Lean vector `![B1, B2, B3, Y, W1, W2, W3]` has seven distinct entries. □
⟨2⟩3. $C_a = Q_1$.
PROOF: Use the fixed list in ⟨2⟩2. In zero-based indices, the named blocks occupy $0, 1, 2$, Y occupies 3, and W_1, W_2, W_3 occupy $4, 5, 6$ by ⟨2⟩1. For each of the 49 pairs, the corresponding entry of ⟨1⟩20 equals the displayed entry of Q_1 in Definition (the C19 matrices): the diagonal is 10, triangle and within-W entries are 3, the three entries from Y to B_1, B_2, B_3 are $(3, -4, -4)$, each W_i has $(-4, 3, 3)$ to those blocks, and all Y, W_i entries are -4. These cases and their reverses exhaust the entries. □
⟨2⟩4. $\det C_a = 0$.
PROOF: Apply Lemma (principal matrices on labelled subsets) to ⟨2⟩2, whose domain has size seven. □
⟨2⟩5. Q.E.D.
PROOF: Steps ⟨2⟩3 and ⟨2⟩4 contradict Lemma (the two nonsingular matrices) at $j = 1$. □
⟨1⟩22. ASSUME: $j = 2$.
PROVE: FALSE.
⟨2⟩1. PICK Y_1, Y_2, W_1, W_2 with $\mathcal{U}_0 = \{Y_1, Y_2\}$, $\mathcal{V}_1 = \{W_1, W_2\}$, $Y_1 \neq Y_2$, and $W_1 \neq W_2$.
PROOF: The two sizes in ⟨1⟩18 are both two by ⟨1⟩. Apply `Finset.card_eq_two` from F1 to each set. □
⟨2⟩2. Put $a \triangleq (B_1, B_2, B_3, Y_1, Y_2, W_1, W_2) : \mathrm{Fin}\, 7 \to \mathcal{B}$. Then a is injective.
PROOF: The initial blocks are distinct by ⟨0⟩:2, ⟨1⟩6, and ⟨1⟩15. The disjointness in ⟨1⟩18 and the two inequalities in ⟨2⟩1 separate all remaining pairs. This is the literal Lean vector `![B1, B2, B3, Y1, Y2, W1, W2]`. □
⟨2⟩3. $C_a = Q_2$.
PROOF: The list in ⟨2⟩2 puts B_1, B_2, B_3 at indices $0, 1, 2$, Y_1, Y_2 at $3, 4$, and W_1, W_2 at $5, 6$, using ⟨2⟩1. For each of the 49 pairs, ⟨1⟩20 matches Q_2 in Definition (the C19 matrices): the diagonal is 10, the triangle, Y_1, Y_2, and W_1, W_2 entries are 3, each Y_i has $(3, -4, -4)$ to the triangle, each W_i has $(-4, 3, 3)$, and the four Y_i, W_h entries are -4. The reverse entries agree by symmetry. □
⟨2⟩4. $\det C_a = 0$.
PROOF: Lemma (principal matrices on labelled subsets) applies to ⟨2⟩2 with domain size seven. □
⟨2⟩5. Q.E.D.
PROOF: Steps ⟨2⟩3 and ⟨2⟩4 contradict Lemma (the two nonsingular matrices) at $j = 2$. □
⟨1⟩23. $j = 0$; B_3 is adjacent to every member of $\mathcal{U}$ and to no member of $\mathcal{M} \setminus \{B_3\}$.
PROOF: The integer bound $0 \leq j \leq 2$ of ⟨1⟩16 and the exclusions ⟨1⟩21, ⟨1⟩22 force zero. The counts in ⟨1⟩16, ⟨1⟩17 are then four and zero; use F1. □
⟨1⟩24. $\mathcal{M} \setminus \{B_3\}$ is a clique of size four.
⟨2⟩1. $|\mathcal{M} \setminus \{B_3\}| = 4$.
PROOF: Use $|\mathcal{M}| = 5$, $B_3 \in \mathcal{M}$ from ⟨1⟩5, ⟨1⟩15, and deletion of a singleton in F1. □
⟨2⟩2. ASSUME: $X, Y \in \mathcal{M} \setminus \{B_3\}$ are distinct and nonadjacent.
PROVE: FALSE.
PROOF: The three distinct vertices B_3, X, Y are neighbours of B_1 by ⟨1⟩1 and ⟨1⟩15; B_3 has no edges to X, Y by ⟨1⟩23. With ⟨2⟩ they give a claw, contradicting Lemma (no induced claw). □
⟨2⟩3. Q.E.D.

PROOF: Step ⟨2⟩1 gives the size and ⟨2⟩2 gives every required edge. □

⟨1⟩25. Put $S_1 \triangleq \{B_1, B_2\} \cup (\mathcal{M} \setminus \{B_3\})$. Then $S_1 \in \mathcal{S}$ and $B_1, B_2 \in S_1$.

⟨2⟩1. $|S_1| = 6$.

PROOF: The union is disjoint by ⟨1⟩6, and ⟨1⟩24 gives $2 + 4 = 6$. □

⟨2⟩2. B_1, B_2 are adjacent.

PROOF: Hypothesis ⟨0⟩:2. □

⟨2⟩3. Every member of $\mathcal{M} \setminus \{B_3\}$ is adjacent to each of B_1, B_2.

PROOF: The definition of $\mathcal{M}$ in ⟨1⟩1. □

⟨2⟩4. Distinct members of $\mathcal{M} \setminus \{B_3\}$ are adjacent.

PROOF: Step ⟨1⟩24. □

⟨2⟩5. Q.E.D.

PROOF: The three pair types in ⟨2⟩2–⟨2⟩4 exhaust pairs in S_1. With ⟨2⟩1 this is membership in $\mathcal{S}$, by Definition (the block graph and all six-cliques); the two containments follow from the displayed union. □

⟨1⟩26. Put $S_2 \triangleq \{B_1, B_3\} \cup \mathcal{U}$. Then $S_2 \in \mathcal{S}$.

⟨2⟩1. $|S_2| = 6$.

PROOF: The two-element set is disjoint from $\mathcal{U}$ by ⟨1⟩6 and ⟨1⟩15; its union has size $2+4 = 6$ by ⟨1⟩5. □

⟨2⟩2. $B_1 \sim B_3$.

PROOF: Step ⟨1⟩15 puts $B_3 \in \mathcal{M}$. □

⟨2⟩3. Every member of $\mathcal{U}$ is adjacent to B_1.

PROOF: Step ⟨1⟩2. □

⟨2⟩4. Every member of $\mathcal{U}$ is adjacent to B_3.

PROOF: Step ⟨1⟩23. □

⟨2⟩5. Every pair of distinct members of $\mathcal{U}$ is adjacent.

PROOF: Step ⟨1⟩10. □

⟨2⟩6. Q.E.D.

PROOF: Steps ⟨2⟩2–⟨2⟩5 cover all pair types. Together with ⟨2⟩1, they are the predicate of Definition (the block graph and all six-cliques). □

⟨1⟩27. $S_1 \cap S_2 = \{B_1\}$.

PROOF: In ⟨1⟩25 and ⟨1⟩26, the parts $\{B_2\}$, $\mathcal{M} \setminus \{B_3\}$, $\{B_3\}$, and $\mathcal{U}$ are pairwise disjoint by ⟨1⟩6 and ⟨1⟩15; only B_1 is shared. □

⟨1⟩28. $S_1 \cup S_2 = \{B_1\} \cup \Gamma(B_1)$.

PROOF: The two unions in ⟨1⟩25, ⟨1⟩26 combine to $\{B_1, B_2\} \cup \mathcal{M} \cup \mathcal{U}$, since $B_3 \in \mathcal{M}$. Use the first neighbourhood decomposition of ⟨1⟩8. □

⟨1⟩29. ASSUME: $K \in \mathcal{S}$ and $B_1, B_2 \in K$.
PROVE: $K = S_1$.

⟨2⟩1. $K \setminus \{B_1, B_2\} \subseteq \mathcal{M}$ and $|K \setminus \{B_1, B_2\}| = 4$.

PROOF: Every remaining vertex is adjacent to both initial blocks by Definition (the block graph and all six-cliques). The two distinct blocks are in the six-element set K, so F1 gives size four. □

⟨2⟩2. $B_3 \notin K$.

⟨3⟩1. ASSUME: $B_3 \in K$.
PROVE: FALSE.

PROOF: By ⟨2⟩1, removing B_3 leaves three vertices in $\mathcal{M} \setminus \{B_3\}$, thus at least one, by F1. The clique K makes it adjacent to B_3, contradicting ⟨1⟩23. □

⟨3⟩2. Q.E.D.

PROOF: Step ⟨3⟩1 proves the negation. □

⟨2⟩3. $K \setminus \{B_1, B_2\} = \mathcal{M} \setminus \{B_3\}$.
PROOF: Steps ⟨2⟩1, ⟨2⟩2 give inclusion; both sets have size four by ⟨1⟩24. Apply the subset and cardinality criterion in F1. □
⟨2⟩4. Q.E.D.
PROOF: Reinsert the two members B_1, B_2 into ⟨2⟩3; the resulting union is S_1 by ⟨1⟩25. □
⟨1⟩30. Q.E.D.
PROOF: Steps ⟨1⟩25, ⟨1⟩26, ⟨1⟩27, ⟨1⟩28, and ⟨1⟩29 give the stated witnesses and uniqueness. □

Corollary (the unique six-clique through an edge).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $B_1 \sim B_2$.
PROVE: There is a unique $S \in \mathcal{S}$ containing B_1, B_2. In particular, if $K, L \in \mathcal{S}$ both contain B_1, B_2, then $K = L$.

PROOF SKETCH: Choose one local witness and compare every other six-clique through the edge with that same witness.
⟨1⟩1. PICK S_1, S_2 with the conclusions of Lemma (splitting a neighbourhood) for B_1, B_2.
PROOF: Apply that lemma to ⟨0⟩:2. □
⟨1⟩2. There is a unique $S \in \mathcal{S}$ containing B_1, B_2.
PROOF: Step ⟨1⟩1 gives the witness S_1 and states that every such S equals S_1. □
⟨1⟩3. ASSUME: $K, L \in \mathcal{S}$ and $B_1, B_2 \in K \cap L$.
PROVE: $K = L$.
PROOF: The uniqueness clause of ⟨1⟩1 gives $K = S_1$ and $L = S_1$. Transitivity gives $K = L$. □
⟨1⟩4. Q.E.D.
PROOF: Steps ⟨1⟩2 and ⟨1⟩3. □

Corollary (exactly two six-cliques through a block).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $B_1 \in \mathcal{B}$.
PROVE: There are distinct $S_1, S_2 \in \mathcal{S}$ with $\{S \in \mathcal{S} : B_1 \in S\} = \{S_1, S_2\}$ and $S_1 \cap S_2 = \{B_1\}$. In particular the filtered family has size two.

PROOF SKETCH: We will use one local split, then apply the stated adjacent-pair uniqueness lemma to each possible six-clique.
⟨1⟩1. PICK $B_2 \in \Gamma(B_1)$.
PROOF: Lemma (block intersections) gives ten neighbours; positive cardinality in F1 gives a witness. □
⟨1⟩2. PICK S_1, S_2 with all the conclusions of Lemma (splitting a neighbourhood) for B_1, B_2.
PROOF: Apply that lemma using ⟨1⟩1. □
⟨1⟩3. $S_1 \neq S_2$, and both contain B_1.
PROOF: Their intersection is $\{B_1\}$ by ⟨1⟩2. Equality would give a singleton of size six, contrary to Definition (the block graph and all six-cliques). □
⟨1⟩4. ASSUME: $K \in \mathcal{S}$ and $B_1 \in K$.
PROVE: $K = S_1$ or $K = S_2$.
⟨2⟩1. PICK $X \in K \setminus \{B_1\}$.
PROOF: This set has size five by ⟨1⟩, Definition (the block graph and all six-cliques), and F1. □
⟨2⟩2. $B_1 \sim X$ and $X \in (S_1 \setminus \{B_1\}) \cup (S_2 \setminus \{B_1\})$.

PROOF: The clique K in $\langle 1\rangle$ and $\langle 2\rangle 1$ give adjacency. The covering conclusion of $\langle 1\rangle 2$ gives the indicated membership. □

$\langle 2\rangle 3$. CASE: $X \in S_1$.
$K = S_1$.
PROOF: Both six-cliques contain the adjacent pair B_1, X by $\langle 1\rangle$, $\langle 1\rangle 3$, and $\langle 2\rangle 2$. Apply the uniqueness conclusion of Corollary (the unique six-clique through an edge) to this pair. □

$\langle 2\rangle 4$. CASE: $X \in S_2$.
$K = S_2$.
PROOF: The cliques K, S_2 contain B_1, X by $\langle 1\rangle$, $\langle 1\rangle 3$, and $\langle 2\rangle 2$. Corollary (the unique six-clique through an edge) applied to B_1, X identifies them. □

$\langle 2\rangle 5$. Q.E.D.
PROOF: Step $\langle 2\rangle 2$ makes $\langle 2\rangle 3$, $\langle 2\rangle 4$ exhaustive. □

$\langle 1\rangle 5$. Q.E.D.
PROOF: Step $\langle 1\rangle 4$ gives containment of the filtered family in $\{S_1, S_2\}$; $\langle 1\rangle 3$ gives the reverse containment and size two. The intersection is part of $\langle 1\rangle 2$. □

1.6. **The global family.**

Lemma (seven six-cliques).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: $|\mathcal{S}| = 7$.

PROOF SKETCH: Count incidences between block labels and six-cliques.
$\langle 1\rangle 1$. There are $6|\mathcal{S}|$ pairs (B, S) with $B \in S \in \mathcal{S}$.
PROOF: Every member of $\mathcal{S}$ has size six by Definition (the block graph and all six-cliques); apply F1. □
$\langle 1\rangle 2$. There are 42 such pairs.
PROOF: Each of the 21 block labels belongs to two members of $\mathcal{S}$ by Corollary (exactly two six-cliques through a block) and Definition (labelled designs); use F1. □
$\langle 1\rangle 3$. Q.E.D.
PROOF: Equate $\langle 1\rangle 1$ and $\langle 1\rangle 2$, then divide $6|\mathcal{S}| = 42$ by six. □

Lemma (intersections of six-cliques).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $S_0, S_1 \in \mathcal{S}$ and $S_0 \neq S_1$.
PROVE: $|S_0 \cap S_1| = 1$.

PROOF SKETCH: Any two distinct cliques share at most one block. The other clique through each of the six blocks of S_0 then enumerates all six cliques different from S_0.
$\langle 1\rangle 1$. For all distinct $S, T \in \mathcal{S}$, $|S \cap T| \leq 1$.
$\langle 2\rangle 1$. ASSUME: $S, T \in \mathcal{S}$, $S \neq T$, and distinct $B, B' \in S \cap T$.
PROVE: FALSE.
PROOF: $B \sim B'$ by the clique predicate in Definition (the block graph and all six-cliques). Corollary (the unique six-clique through an edge) identifies S, T through this edge, contradicting $S \neq T$. □
$\langle 2\rangle 2$. Q.E.D.
PROOF: The cardinality-at-most-one criterion of F1 applies to $\langle 2\rangle 1$. □
$\langle 1\rangle 2$. For every $B \in S_0$ there is a unique $f(B) \in \mathcal{S} \setminus \{S_0\}$ containing B.
PROOF: Corollary (exactly two six-cliques through a block) supplies exactly two cliques through B, one of which is S_0 by $\langle 0\rangle$:2. Deleting that one leaves a singleton by F1. □

⟨1⟩3. $f : S_0 \to \mathcal{S} \setminus \{S_0\}$ is injective.
PROOF: If $f(B) = f(B')$ with $B \neq B'$, then S_0 and $f(B) \neq S_0$ share two blocks by ⟨1⟩2, contradicting ⟨1⟩1. □
⟨1⟩4. f is surjective.
PROOF: Its domain has size six by Definition (the block graph and all six-cliques); its codomain has size $7 - 1 = 6$ by Lemma (seven six-cliques) and ⟨0⟩:2. The finite bijectivity criterion of F1 applies to ⟨1⟩3. □
⟨1⟩5. $S_0 \cap S_1$ is nonempty.
PROOF: By ⟨0⟩:2 and ⟨1⟩4, choose $B \in S_0$ with $f(B) = S_1$. Step ⟨1⟩2 gives $B \in S_1$. □
⟨1⟩6. Q.E.D.
PROOF: Combine the lower bound from ⟨1⟩5 with ⟨1⟩1, using F1. □

Lemma (points in a six-clique).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality and D is a design with parameters $(15, 5, 21, 7, 2)$.
2. $S_0 \in \mathcal{S}$.
PROVE: For every $p \in P$, $|\{B \in S_0 : p \in B\}| = 2$.

PROOF SKETCH: Two moments of the point multiplicities give a sum of squares equal to zero.
⟨1⟩1. LET: $m(p) \triangleq |\{B \in S_0 : p \in B\}|$.
Then $\sum_{p \in P} m(p) = 30$.
PROOF: Count (p, B) with $B \in S_0$ and $p \in B$. There are six labels, each with five points, by Definitions (the block graph and all six-cliques), (labelled designs), and F1. □
⟨1⟩2. $\sum_{p \in P} m(p)(m(p) - 1) = 30$ in $\mathbb{Z}$.
PROOF: Count (p, B, B') with distinct $B, B' \in S_0$ and $p \in B \cap B'$. Each of the $6 \cdot 5$ ordered pairs meets in exactly one point by Definition (the block graph and all six-cliques). Counting by p gives the sum, using F1. □
⟨1⟩3. $\sum_{p \in P} (m(p) - 2)^2 = 0$ in $\mathbb{Z}$.
PROOF: Integer expansion gives

$$\begin{aligned} \sum_p (m(p) - 2)^2 &= \sum_p m(p)(m(p) - 1) - 3 \sum_p m(p) + 4|P| && \text{[ring arithmetic]} \\ &= 30 - 90 + 60 && \text{[⟨1⟩1, ⟨1⟩2, (labelled designs)]} \\ &= 0 && \text{[integer arithmetic].} \end{aligned}$$

□
⟨1⟩4. For every $p \in P$, $(m(p) - 2)^2 = 0$.
PROOF: Each integer square is nonnegative; apply the zero-sum criterion of F1 to ⟨1⟩3. □
⟨1⟩5. Q.E.D.
PROOF: An integer square vanishes only at zero, so ⟨1⟩4 gives $m(p) - 2 = 0$ and $m(p) = 2$ for every p. □

1.7. The labelled extension and the matrix it supplies.

Definition (clique incidence and extended blocks).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
LET: $T \in \mathrm{Mat}_{\mathcal{B} \times \mathcal{S}}(\mathbb{Z})$ have $T(B, S) = 1$ for $B \in S$ and 0 otherwise. Put $P^* \triangleq P \sqcup \mathcal{S}$ and $\mathcal{B}^* \triangleq \mathrm{Option}(\mathcal{B})$.

Denote their injections by $\iota : P \to P^*$ and $\eta : \mathcal{S} \to P^*$, the old labels by some$(B)$, and the new label by none. Define the block function $F : \mathcal{B}^* \to \mathrm{Finset}(P^*)$ by

$$F(\mathrm{some}(B)) \triangleq \iota(B) \cup \{\eta(S) : S \in \mathcal{S},\ B \in S\},$$
$$F(\mathrm{none}) \triangleq \{\eta(S) : S \in \mathcal{S}\}.$$

Here $\iota(B)$ is the image of the point set of B. Define $N^* \triangleq D.\mathrm{hallConnorIncMatrix} \in \mathrm{Mat}_{P^* \times \mathcal{B}^*}(\mathbb{Z})$ by the four entry formulas

$$N^*(\iota(p), \mathrm{some}(B)) \triangleq N(p, B), \quad N^*(\iota(p), \mathrm{none}) \triangleq 0,$$
$$N^*(\eta(S), \mathrm{some}(B)) \triangleq T(B, S), \quad N^*(\eta(S), \mathrm{none}) \triangleq 1.$$

Lemma (parameters of the extended blocks).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: For the types and block function of Definition (clique incidence and extended blocks),

$$|P^*| = 22, \qquad |\mathcal{B}^*| = 22, \qquad \forall A \in \mathcal{B}^*,\ |F(A)| = 7,$$
$$\forall x \in P^*,\ |\{A : x \in F(A)\}| = 7,$$
$$\forall x, y \in P^*,\ x \neq y \Rightarrow |\{A : x, y \in F(A)\}| = 2.$$

PROOF SKETCH: We will verify every cardinality and each of the three unordered types of point pair. The disjoint sum distinguishes new points from old points even if their underlying representations agree.

⟨1⟩1. $|P^*| = 22$ and $|\mathcal{B}^*| = 22$.
PROOF: Definition (clique incidence and extended blocks) gives the sums $|P| + |\mathcal{S}| = 15 + 7$ and $|\mathcal{B}| + 1 = 21 + 1$, using Definition (labelled designs), Lemma (seven six-cliques), and `Fintype.card_sum` and `Fintype.card_option` in F1. □

⟨1⟩2. For every $B \in \mathcal{B}$, $|F(\mathrm{some}(B))| = 7$.
PROOF: There are five old points by Definition (labelled designs) and two new points by Corollary (exactly two six-cliques through a block). Both ι and η are injective, and their images are disjoint by Definition (clique incidence and extended blocks). Apply `Finset.card_image_of_injective` and the disjoint-union count in F1 to get $5 + 2 = 7$. □

⟨1⟩3. $|F(\mathrm{none})| = 7$.
PROOF: The map η is injective, and Lemma (seven six-cliques) gives seven members of $\mathcal{S}$; use Definition (clique incidence and extended blocks) and `Finset.card_image_of_injective` in F1. □

⟨1⟩4. Each old point $\iota(p)$ lies in seven extended blocks.
PROOF: Definition (clique incidence and extended blocks) preserves all incidences of p with old labels and puts no old point in $F(\mathrm{none})$. Its old replication number is seven by Definition (labelled designs). □

⟨1⟩5. Each new point $\eta(S)$ lies in seven extended blocks.
PROOF: Definition (clique incidence and extended blocks) puts it in exactly the old labels $B \in S$, of which there are six by Definition (the block graph and all six-cliques), and also in the distinct new label none. The count is $6 + 1 = 7$. □

⟨1⟩6. ASSUME: $p, q \in P$ and $p \neq q$.
PROVE: $\iota(p), \iota(q)$ lie together in exactly two extended blocks.
⟨2⟩1. The common old labels are precisely $\{\mathrm{some}(B) : p, q \in B\}$.
PROOF: The old-incidence clause of Definition (clique incidence and extended blocks). □
⟨2⟩2. The new block contains neither old point.

PROOF: The two injections have disjoint images in Definition (clique incidence and extended blocks). □

⟨2⟩3. Q.E.D.

PROOF: Step ⟨2⟩1 has cardinality two by Definition (labelled designs) and ⟨1⟩; ⟨2⟩2 excludes any further label. □

⟨1⟩7. ASSUME: $p \in P$ and $S \in \mathcal{S}$.

PROVE: $\iota(p), \eta(S)$ lie together in exactly two extended blocks.

⟨2⟩1. The common old labels are precisely $\{\text{some}(B) : B \in S,\ p \in B\}$.

PROOF: Use both old-label incidence clauses in Definition (clique incidence and extended blocks). □

⟨2⟩2. The new block does not contain $\iota(p)$.

PROOF: Definition (clique incidence and extended blocks). □

⟨2⟩3. Q.E.D.

PROOF: Lemma (points in a six-clique) gives two labels in ⟨2⟩1; ⟨2⟩2 excludes a contribution from the new label. □

⟨1⟩8. ASSUME: $S \in \mathcal{S}$ and $p \in P$.

PROVE: $\eta(S), \iota(p)$ lie together in exactly two extended blocks.

PROOF: The set of common labels is unchanged when the two membership conditions are interchanged. Apply ⟨1⟩7 to p, S. □

⟨1⟩9. ASSUME: $S, T \in \mathcal{S}$ and $S \neq T$.

PROVE: $\eta(S), \eta(T)$ lie together in exactly two extended blocks.

⟨2⟩1. The common old labels are $\{\text{some}(B) : B \in S \cap T\}$, of size one.

PROOF: Definition (clique incidence and extended blocks) gives this set, and Lemma (intersections of six-cliques) with ⟨1⟩ gives its size. □

⟨2⟩2. Both points lie in the new block.

PROOF: The clause for $F(\text{none})$ in Definition (clique incidence and extended blocks). □

⟨2⟩3. Q.E.D.

PROOF: The old label of ⟨2⟩1 and the new label of ⟨2⟩2 are distinct, and together exhaust the common labels by Definition (clique incidence and extended blocks). □

⟨1⟩10. Q.E.D.

PROOF: Steps ⟨1⟩1–⟨1⟩5 verify the cardinalities, block sizes, and replication numbers. Steps ⟨1⟩6, ⟨1⟩7, ⟨1⟩8, ⟨1⟩9 exhaust ordered distinct point pairs: each point is in exactly one summand. Counting labels uses `Fintype.sum_option`, and splitting point sums uses `Fintype.sum_sum_type`, from F1. These give every asserted equality. □

Definition (the Hall–Connor extension).

ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.

2. D is a design with parameters $(15, 5, 21, 7, 2)$.

Define $D^* \triangleq D$.hallConnorExtension of type

```
Design (Sum P D.sixCliques) (Option B) 22 7 22 7 2
```

with D^*.block $\triangleq D$.hallConnorBlocks $= F$ from Definition (clique incidence and extended blocks). The subtype of members is intended by `D.sixCliques` in the point type. The cardinality and incidence fields are exactly the equalities in Lemma (parameters of the extended blocks); the positivity fields of Definition (labelled designs) are $22 > 1$, $7 > 1$, and $22 > 0$.

Theorem (the Hall–Connor residual identity). [[HC], Theorems 2.1 and 3.2 at $k = 5$.]

ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.

2. D is a design with parameters $(15, 5, 21, 7, 2)$.

PROVE: The design $D^* = D.\mathrm{hallConnorExtension}$ is symmetric, and for every $B \in \mathcal{B}$,

$$D^*.\mathrm{block}(\mathrm{some}(B)) \setminus D^*.\mathrm{block}(\mathrm{none}) = \iota(B).$$

Thus deleting the new block and all its points recovers D label by label.

PROOF SKETCH: The parameter type gives symmetry; the two disjoint point images give the residual identity.

⟨1⟩1. D^* is symmetric.
PROOF: Its parameters in Definition (the Hall–Connor extension) have $v = b = 22$, which is Definition (labelled designs). □

⟨1⟩2. For every B, the asserted block difference is $\iota(B)$.
PROOF: Unfold the block function in Definition (the Hall–Connor extension). In Definition (clique incidence and extended blocks), every added point belongs to $F(\mathrm{none})$ and no old point does. Removing that block from $F(\mathrm{some}(B))$ leaves exactly $\iota(B)$. □

⟨1⟩3. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2. □

Lemma (integral products for the extension).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.
PROVE: All of the following are identities over $\mathbb{Z}$:

$$NT = 2J_{P,\mathcal{S}}, \qquad T^T T = 5I_{\mathcal{S}} + J_{\mathcal{S}}, \qquad N^* N^{*T} = 5I_{P^*} + 2J_{P^*}.$$

PROOF SKETCH: We will expose the four matrix blocks so that the main contradiction can consume the integral identity directly.

⟨1⟩1. $NT = 2J_{P,\mathcal{S}}$.
PROOF: By Definition (clique incidence and extended blocks) and F2, the (p, S) entry counts $B \in S$ with $p \in B$. Lemma (points in a six-clique) gives two. □

⟨1⟩2. $(T^T T)(S, S) = 6$ for every $S \in \mathcal{S}$.
PROOF: This entry counts the six labels in S, by Definitions (the block graph and all six-cliques), (clique incidence and extended blocks), and F2. □

⟨1⟩3. ASSUME: $S, T_0 \in \mathcal{S}$ and $S \neq T_0$.
PROVE: $(T^T T)(S, T_0) = 1$.
PROOF: The entry counts $S \cap T_0$, by Definition (clique incidence and extended blocks) and F2; Lemma (intersections of six-cliques) gives one. □

⟨1⟩4. $T^T T = 5I_{\mathcal{S}} + J_{\mathcal{S}}$.
PROOF: The diagonal and off-diagonal cases ⟨1⟩2 and ⟨1⟩3 are exhaustive and have the asserted entries 6 and 1. □

⟨1⟩5. The old-old block of $N^* N^{*T}$ is $5I_P + 2J_P$.
PROOF: Sum over old labels and the new label using Definition (clique incidence and extended blocks), F2, and `Fintype.sum_option` in F1. The result is $NN^T + 0$, equal to $5I_P + 2J_P$ by Lemma (incidence identities). □

⟨1⟩6. The old-new block of $N^* N^{*T}$ is $2J_{P,\mathcal{S}}$.
PROOF: The two label contributions are NT and zero, by Definition (clique incidence and extended blocks), F2, and `Fintype.sum_option` in F1; use ⟨1⟩1. □

⟨1⟩7. The new-old block of $N^* N^{*T}$ is $2J_{\mathcal{S},P}$.
PROOF: It is the transpose of the old-new block, since a matrix times its transpose is symmetric by F2; apply ⟨1⟩6. □

⟨1⟩8. The new-new block of $N^* N^{*T}$ is $5I_{\mathcal{S}} + 2J_{\mathcal{S}}$.

PROOF: Old labels contribute $T^T T$ and the new label contributes $J_{\mathcal{S}}$, by Definition (clique incidence and extended blocks) and F2, using `Fintype.sum_option` in F1. Step ⟨1⟩4 gives their sum. □

⟨1⟩9. Q.E.D.

PROOF: Steps ⟨1⟩5–⟨1⟩8 give the four blocks of $5I_{P^*} + 2J_{P^*}$, because distinct summands never give a diagonal entry. Together with ⟨1⟩1 and ⟨1⟩4, they prove all assertions. □

Theorem (Hall–Connor, matrix form).

ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 5, 21, 7, 2)$.

PROVE: There is $M \in \mathrm{Mat}_{\mathrm{Fin}\,22 \times \mathrm{Fin}\,22}(\mathbb{Z})$ with $MM^T = 5I_{22} + 2J_{22}$.

PROOF SKETCH: We will reindex both finite types in the integral identity, keeping the summed block index the same in both factors.

⟨1⟩1. $|P^*| = 22$ and $|\mathcal{B}^*| = 22$.

PROOF: Definition (clique incidence and extended blocks) gives disjoint sums of sizes $15 + 7$ and $21 + 1$, by Definition (labelled designs), Lemma (seven six-cliques), and F1. □

⟨1⟩2. PICK equivalences $a : \mathrm{Fin}\,22 \to P^*$ and $b : \mathrm{Fin}\,22 \to \mathcal{B}^*$.

PROOF: Apply the enumeration fact in F1 to ⟨1⟩1. □

⟨1⟩3. LET: $M(i, j) \triangleq N^*(a(i), b(j))$.
This is an integral matrix.

PROOF: Every entry of N^* is an integer by Definition (clique incidence and extended blocks); composition preserves its scalar type. □

⟨1⟩4. $(MM^T)(i, j) = (N^* N^{*T})(a(i), a(j))$ for all i, j.

PROOF: Both factors use the bijection b on the summed label index. Apply `Matrix.transpose_submatrix` and `Matrix.submatrix_mul_equiv` from F2 to ⟨1⟩3, using ⟨1⟩2. □

⟨1⟩5. $MM^T = 5I_{22} + 2J_{22}$.

PROOF: Lemma (integral products for the extension) and ⟨1⟩4 give entries $5\,1_{a(i)=a(j)} + 2$. The injection a in ⟨1⟩2 makes $a(i) = a(j)$ equivalent to $i = j$, so these are exactly the asserted entries. □

⟨1⟩6. Q.E.D.

PROOF: The witness ⟨1⟩3 and its equation ⟨1⟩5 prove the existential conclusion. □

1.8. The determinant obstruction and the conclusion.

Lemma (an odd power of five).

ASSUME: $m \in \mathbb{N}$ and $d \in \mathbb{Z}$.

PROVE: $d^2 \neq 49 \cdot 5^{2m+1}$. In particular $d^2 \neq 49 \cdot 5^{21}$.

PROOF SKETCH: We will induct on m, universally quantifying the integer at each stage so that the induction hypothesis applies to $d/5$.

⟨1⟩1. For every $m \in \mathbb{N}$ and every $d \in \mathbb{Z}$, $d^2 \neq 49 \cdot 5^{2m+1}$.

⟨2⟩1. ASSUME: $d \in \mathbb{Z}$ and $d^2 = 49 \cdot 5$.
PROVE: FALSE.

⟨3⟩1. PICK $e \in \mathbb{Z}$ with $d = 5e$.

PROOF: The equality in ⟨2⟩ gives $5 \mid d^2$; F5 gives $5 \mid d$. □

⟨3⟩2. $5e^2 = 49$.

PROOF: Substitute ⟨3⟩1 into ⟨2⟩ and cancel the nonzero integer factor five. □

⟨3⟩3. Q.E.D.

PROOF: Step ⟨3⟩2 would make 49 divisible by five, whereas $49 = 5 \cdot 9 + 4$. □

⟨2⟩2. ASSUME: 1. $m \in \mathbb{N}$ and $e^2 \neq 49 \cdot 5^{2m+1}$ for every $e \in \mathbb{Z}$.

2. $d \in \mathbb{Z}$ and $d^2 = 49 \cdot 5^{2m+3}$.

PROVE: FALSE.

⟨3⟩1. PICK $e \in \mathbb{Z}$ with $d = 5e$.

PROOF: The exponent $2m+3$ is positive, so ⟨2⟩:2 gives $5 \mid d^2$. Apply F5. □

⟨3⟩2. $e^2 = 49 \cdot 5^{2m+1}$.

PROOF: Substitute ⟨3⟩1 into ⟨2⟩:2, use $5^{2m+3} = 25 \cdot 5^{2m+1}$, and cancel $25 \neq 0$. □

⟨3⟩3. Q.E.D.

PROOF: Apply ⟨2⟩:1 to the integer e in ⟨3⟩2. □

⟨2⟩3. Q.E.D.

PROOF: Natural induction using ⟨2⟩1 and ⟨2⟩2, since $2(m+1)+1 = 2m+3$. □

⟨1⟩2. Q.E.D.

PROOF: Step ⟨1⟩1 proves the general assertion; its instance $m = 10$ gives exponent 21. □

Theorem (the Bruck–Ryser–Chowla instance).

ASSUME: $M \in \mathrm{Mat}_{\mathrm{Fin}\,22 \times \mathrm{Fin}\,22}(\mathbb{Z})$.

PROVE: $MM^T \neq 5I_{22} + 2J_{22}$.

PROOF SKETCH: We will take determinants over $\mathbb{Q}$, retaining the integer determinant to apply the descent at five.

⟨1⟩1. ASSUME: $MM^T = 5I_{22} + 2J_{22}$.

PROVE: FALSE.

⟨2⟩1. Put $d \triangleq \det_{\mathbb{Z}} M$. Then $d \in \mathbb{Z}$ and $\det_{\mathbb{Q}}(M_{\mathbb{Q}}) = (d : \mathbb{Q})$.

PROOF: The integer determinant is an integer by its scalar type; the cast identity is `Int.cast_det` in F4. □

⟨2⟩2. $(d : \mathbb{Q})^2 = 49 \cdot 5^{21}$.

PROOF: Cast ⟨1⟩ to $\mathbb{Q}$ using F2. Since $5 \neq 0$ in $\mathbb{Q}$, the determinant lemma applies, giving

$$\begin{aligned}
(d : \mathbb{Q})^2 &= \det(M_{\mathbb{Q}} M_{\mathbb{Q}}^T) && [\langle 2\rangle 1,\ \text{F4}] \\
&= \det(5I_{22} + 2J_{22}) && [\langle 1\rangle] \\
&= 5^{21}(5 + 22 \cdot 2) && [\text{(a scalar matrix plus a constant matrix)}] \\
&= 49 \cdot 5^{21} && [\text{rational arithmetic}].
\end{aligned}$$

□

⟨2⟩3. $d^2 = 49 \cdot 5^{21}$ in $\mathbb{Z}$.

PROOF: The cast $\mathbb{Z} \to \mathbb{Q}$ is injective and preserves sums and products; apply it to ⟨2⟩2. □

⟨2⟩4. Q.E.D.

PROOF: Lemma (an odd power of five) contradicts ⟨2⟩3. □

⟨1⟩2. Q.E.D.

PROOF: Step ⟨1⟩1 proves the required negation. □

Theorem (the parameter contradiction).

ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.

2. D is a design with parameters $(15, 5, 21, 7, 2)$.

PROVE: FALSE.

PROOF SKETCH: The matrix supplied by the embedding is forbidden by the obstruction.

⟨1⟩1. PICK an integer matrix M of order 22 with $MM^T = 5I_{22} + 2J_{22}$.

PROOF: Apply Theorem (Hall–Connor, matrix form) to D. □

⟨1⟩2. Q.E.D.

PROOF: Apply Theorem (the Bruck–Ryser–Chowla instance) to the witness and equation in ⟨1⟩1. □

Corollary (nonexistence of a 2-$(15, 5, 2)$ design).

PROVE: The type

```
Design (Fin 15) (Fin 21) 15 5 21 7 2
```

is empty. Moreover, for every finite point type P of size 15 and every finite label type $\mathcal{B}$, both with decidable equality, there is no function $F : \mathcal{B} \to \mathrm{Finset}(P)$ whose blocks have size five and whose distinct point pairs lie in exactly two labels.

PROOF SKETCH: We will state both the bundled and unbundled forms.

⟨1⟩1. The specified design type is empty.
PROOF: Every inhabitant would give FALSE by Theorem (the parameter contradiction) with $P = \mathrm{Fin}\,15$, $\mathcal{B} = \mathrm{Fin}\,21$. □

⟨1⟩2. ASSUME: $P, \mathcal{B}, F$ have the properties in the second assertion.
PROVE: FALSE.
⟨2⟩1. PICK D : `Design P B 15 5 21 7 2` with $D.\mathrm{block}(B) = F(B)$ for every $B \in \mathcal{B}$.
PROOF: Apply Lemma (the forced parameters) to ⟨1⟩. □
⟨2⟩2. Q.E.D.
PROOF: Apply Theorem (the parameter contradiction) to ⟨2⟩1. □

⟨1⟩3. Q.E.D.
PROOF: Steps ⟨1⟩1 and ⟨1⟩2. □

Definition (complementary labelled blocks).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters (v, k, b, r, λ).
3. $2 \le v - k$.
Define D.compl with the same point and label types by

$$D.\mathrm{compl} : \texttt{Design P B v (v-k) b (b-r) (b+lam-2*r)},$$

$$D.\mathrm{compl.block}(B) \triangleq P \setminus D(B).$$

The differences are natural differences. No injectivity of the block function is assumed. The proof below constructs its fields.

PROOF SKETCH: We will give the complement constructor its full parameter type. The bound on the new block size is exactly the condition needed by Definition (labelled designs).

⟨1⟩1. Every complementary block has size $v - k$.
PROOF: Its old block is a subset of P with size k; subtract from $|P| = v$ using Definition (labelled designs) and the finite-difference formula of F1. □

⟨1⟩2. Every point belongs to $b - r$ complementary labels.
PROOF: The r old labels through it form a subset of the b labels. The other labels are exactly its complementary incidences, by Definition (labelled designs) and F1. □

⟨1⟩3. **[new]**
ASSUME: $p, q \in P$ and $p \neq q$.
PROVE: Exactly $b + \lambda - 2r$ complementary labels contain p, q.
⟨2⟩1. Let A_p, A_q be the sets of old incident labels of p, q. Then $|A_p \cup A_q| + \lambda = 2r$.
PROOF: Both sets have size r and their intersection has size λ, by ⟨1⟩ and Definition (labelled designs); use the union-intersection identity in F1. □
⟨2⟩2. If c counts labels containing neither point, then $c + 2r = b + \lambda$.
PROOF: The complement of $A_p \cup A_q$ among labels has size c, so $c + |A_p \cup A_q| = b$ by F1. Add ⟨2⟩1. □
⟨2⟩3. Q.E.D.
PROOF: Step ⟨2⟩2 gives $2r \le b + \lambda$ and $c = b + \lambda - 2r$ in $\mathbb{N}$. These are the common complementary labels by the block definition. This is the ordinary proof's inclusion-exclusion argument stated for general parameters, so the constructor has a precise type. □

⟨1⟩4. Q.E.D.
PROOF: The types keep cardinalities v, b; steps ⟨1⟩1, ⟨1⟩2, and ⟨1⟩3 give the three incidence fields. The new block size is greater than one by ⟨0⟩:3; $v > 1, b > 0$ come from the old design in Definition (labelled designs). □

Corollary (the complementary parameters).
ASSUME: 1. $P, \mathcal{B}$ are finite types with decidable equality.
2. D is a design with parameters $(15, 10, 21, 14, 9)$.
PROVE: FALSE. Thus `Design (Fin 15) (Fin 21) 15 10 21 14 9` is empty.

PROOF SKETCH: Complementing every labelled block produces the forbidden parameters.
⟨1⟩1. Put $D' \triangleq D$.compl. Then D' : `Design P B 15 5 21 7 2`.
PROOF: Apply the constructor of Definition (complementary labelled blocks) to ⟨0⟩:2, with $2 \le 15 - 10 = 5$. Its parameters reduce to $15 - 10 = 5$, $21 - 14 = 7$, and $21 + 9 - 2 \cdot 14 = 2$. These are definitionally equal numerals in Lean. □
⟨1⟩2. Q.E.D.
PROOF: Apply Theorem (the parameter contradiction) to ⟨1⟩1. Since the argument applies to arbitrary finite $P, \mathcal{B}$, it also excludes an inhabitant on Fin 15 and Fin 21. □

1.9. **Interfaces, provenance, and checks.** The statements above make the design-to-matrix bridge explicit: `Design.hall_connor` returns the integer matrix equation that `Matrix.bruck_ryser_chowla_twenty_two` negates. The constructor `Design.hallConnorExtension` has the separately named theorem `Design.hallConnorExtension_residual`, with point type `Sum P D.sixCliques` and label type `Option B`. Here `D.sixCliques` denotes the subtype of members when used as a type. The existential matrix theorem does not need a general residual-design API.

No mathematical repair of the ordinary proof is needed. We have made its moments, entry identities, principal-matrix bounds, and clique incidence counts separately citable, put the determinant lemma before its first use, and expanded the local and extension cases. These are changes of proof structure. The ordinary proof already proves the clique property before counting eight edges, already defines the global family as all six-cliques, and already uses rank to exclude Q_1, Q_2. Those repairs of the source are retained here. Neither the unused rational determinant-sign remark nor the unused source-local patterns are dependencies. Three steps are tagged **[new]**: using the rank-one determinant lemma, taking the determinant of each certified inverse identity, and stating the complement count for general parameters to give `Design.compl` an exact interface. They retain the same nonexistence argument.

All direct dependencies have local labels prefixed by `hc:`; there are no deferred blueprint labels. The names in `\lean` are declarations of the Lean formalization, except for the Mathlib API cited explicitly with paths. Mathlib was inspected at commit
`082e2d37e8b0463410cdb532e111cd43d5a66174`
(4 September 2026), the version pinned in `lake-manifest.json`. Its rank identities apply over $\mathbb{Q}$; the analytic theorem about nonnegative determinants over an `RCLike` field is unnecessary. The determinant obstruction uses only the even-order instance, proved here without a quadratic-form local-global theorem.

The six displayed matrices are those of the ordinary proof. The exact checks of the ordinary proof, in `cgss/proofs/hallconnor/checks/`, check all six matrices, the two quadratic values, and every entry of both inverse certificates; `check_t7.py` there is the $T(7)$ control, and the copies in `blueprint/checks/hallconnor/` run the same checks on the matrices of the blueprint. The control exercises all 210 ordered adjacent pairs; no existing design can exercise the hypothetical incidence structure. These checks do not establish the quantified counting and clique arguments, which are provided by the proofs above.

References

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